

Nonconformity Measures and Ensemble Strategies

An Analysis of Conformal Predictor Efficiency and Validity

Henrik Linusson

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Academic dissertation for the Degree of Doctor of Philosophy in Computer and Systems Sciences at Stockholm University to be publicly defended on Monday 14 June 2021 at 13.00 in online via Zoom, public link is available at the department website.

Abstract

Conformal predictors are a family of predictive models that associate with each of their predictions a measure of confidence, enabling them to provide quantitative information about their own trustworthiness. In risk-laden machine learning applications, where bad predictions may lead to economic loss, personal injury, or worse, such inherent quality control appears highly beneficial, if not required. While the foundations of conformal prediction were initially published some twenty years ago, their use, and further development, is still (at the time of writing this thesis) not widespread in the machine learning community, and several open questions remain regarding the proper design and use of conformal prediction systems. In this thesis, we attempt to tackle some of these questions, focusing our attention on three specific characteristics of conformal predictors. First, conformal predictors rely on so-called nonconformity functions, which are mappings from the object space onto the real line, typically based on traditional classification or regression models; here, we investigate properties of the underlying learning algorithm and characteristics of the resulting conformal predictor. Second, conformal predictors output predictions on a form that is distinct from traditional prediction methods, by supplying multi-valued prediction regions with a statistically valid coverage probability; we propose two procedures for post-processing the output from conformal classification models that provide interpretations more closely related to traditional predictive models, while still retaining meaningful confidence information. Finally, we provide contributions relating to the construction of conformal predictor ensembles, illustrating potential issues with existing ensemble procedures, as well as proposing and evaluating an alternative ensemble method.

Keywords: *Data Science, Machine Learning, Conformal Prediction, Classification, Regression.*

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This thesis is dedicated
to my late grandfather,
Knut Valdemar
Svensson. Thank you for
inspiring me to always
be curious.

Abstract

Conformal predictors are a family of predictive models that associate with each of their predictions a measure of confidence, enabling them to provide quantitative information about their own trustworthiness. In risk-laden machine learning applications, where bad predictions may lead to economic loss, personal injury, or worse, such inherent quality control appears highly beneficial, if not required. Although the foundations of conformal prediction were initially published some twenty years ago, their use, and further development, is still (at the time of writing this thesis) not widespread in the machine learning community, and several open questions remain regarding the proper design and use of conformal prediction systems. In this thesis, we attempt to tackle some of these questions, focusing on three specific characteristics of conformal predictors. First, conformal predictors rely on so-called nonconformity functions, which map the object space onto the real line, typically based on traditional classification or regression models. Here, we investigate properties of the underlying learning algorithm and characteristics of the resulting conformal predictor. Second, unlike traditional prediction methods, conformal predictors supply multi-valued prediction regions with a statistically valid coverage probability. We propose two procedures for post-processing the output from conformal classification models that provide interpretations more closely related to traditional predictive models, while still retaining meaningful confidence information. Finally, we provide contributions relating to the construction of conformal predictor ensembles, illustrating potential issues with existing ensemble procedures, as well as proposing and evaluating an alternative ensemble method.

List of Papers

The following papers, referred to in the text by their Roman numerals, are included in this thesis. Reprints were made with permission from the publishers. In each included paper, study design, empirical work and a majority of the writing was done by me.

- I: Henrik Linusson, Ulf Johansson, and Tuve Löfström. Signed-error conformal regression. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 224–236. Springer, 2014b
 - II: Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Efficiency comparison of unstable transductive and inductive conformal classifiers. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 261–270. Springer, 2014a
 - III: Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Reliable confidence predictions using conformal prediction. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 77–88. Springer, 2016
 - IV: Henrik Linusson, Ulf Norinder, Henrik Boström, Ulf Johansson, and Tuve Löfström. On the calibration of aggregated conformal predictors. In *Conformal and probabilistic prediction and applications*, pages 154–173. PMLR, 2017
 - V: Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Classification with reject option using conformal prediction. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 94–105. Springer, 2018
 - VI: Henrik Linusson, Ulf Johansson, and Henrik Boström. Efficient conformal predictor ensembles. *Neurocomputing*, 397:266–278, 2020
-

The following papers also contribute to this thesis, but are not included. My contributions to these publications are primarily limited to assisting in developing core concepts, as well as writing. The method and theory developed in these papers lay a foundation for the out-of-bag calibration framework presented in Paper VI.

1. Ulf Johansson, Henrik Boström, Tuve Löfström, and Henrik Linusson. Regression conformal prediction with random forests. *Machine Learning*, 97(1-2):155–176, 2014
2. Henrik Boström, Henrik Linusson, Tuve Löfström, and Ulf Johansson. Evaluation of a variance-based nonconformity measure for regression forests. In *Symposium on Conformal and Probabilistic Prediction with Applications*, pages 75–89. Springer, 2016
3. Henrik Boström, Henrik Linusson, Tuve Löfström, and Ulf Johansson. Accelerating difficulty estimation for conformal regression forests. *Annals of Mathematics and Artificial Intelligence*, 81(1-2):125–144, 2017

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1. Introduction

The pay's not great, but the work is hard.

—Bernard Black

1.1 Background

Ever since Arthur Samuel [1959] coined the term “machine learning” during his seminal research on self-improving computer checkers programs, researchers have sought to assess the quality of the material learned and retained by such programs. In modern terms, we often speak of the accuracy, precision, the root of the mean squared error, area under the curve—or other increasingly esoteric metrics—of our predictive models, in order to convey some idea of how well they perform. These metrics help us answer questions such as “how often is my model right or wrong?” or “on average, how far is my model from the truth?”, whereas ultimately, perhaps, what we really wish to know is: should we trust our models’ predictions?

The conformal prediction framework [Vovk et al., 2005] provides an alternative take on the traditional method of assessing model quality through various metrics, by providing the means for creating predictive models that can, themselves, provide information about their accuracy and reliability.

When making a prediction for an object whose label is unknown, a traditional predictive model typically provides us with a single label—that which the model deems most likely to be correct—or, in some cases, an *a posteriori* estimation of the distribution of possible labels. In both these cases, we rely on an external assessment of the model, in order to determine the extent to which we should trust it. A hold-out measurement of accuracy or root-mean-square error or whichever error metric we choose provides us with some insight into how well the model performs. These assessments, however, usually display two unfavourable characteristics: they are often biased in that they tend to either underestimate or overestimate the error of the model when applied to a new object with unknown label; and, the quality assessment normally applies on a model level rather than an object level, in that the model quality is expressed

as an average performance over the hold-out set.

A conformal predictor, on the other hand, outputs multi-valued predictions, dubbed *prediction regions*, that express a set or range of possible labels that might be assigned to the object. By carefully designing these prediction regions, the conformal prediction framework is able to provide us with a more explicit sense of prediction quality: the *a priori* probability that the prediction region will contain the true label is known and this probability is valid per object, rather than on an aggregated model level. That is, the amount of trust that we should assign to a prediction provided by a conformal predictor is thus automatically expressed within the prediction itself in terms of a *confidence measure* without the need to rely on external validation.

The true strength of confidence predictions as produced by the conformal prediction framework shows in high-risk applications where making an erroneous prediction might incur a high cost in terms of economic loss, personal injury or even death. Some prime examples of applications of conformal predictors include medical diagnosing [Devetyarov et al., 2012; Gamberman et al., 2008; Lambrou et al., 2011; Nouretdinov et al., 2011; Papadopoulos et al., 2009], virtual screening of drug candidates [Eklund et al., 2012, 2015; Norinder et al., 2014; Svensson et al., 2017, 2018; Toccaceli et al., 2016] and security related monitoring [Cherubin et al., 2015; Laxhammar and Falkman, 2010; Papadopoulos et al., 2018; Wechsler, 2015].

Compared to other frameworks for quantifying uncertainty or producing prediction regions, conformal prediction theory provides several benefits. First, unlike the traditional jackknife [Miller, 1974] and bootstrap [Stine, 1985] procedures, conformal prediction regions are automatically well-calibrated under loose conditions. Second, compared to the Probably Approximately Correct (PAC) learning framework [Valiant, 1984], conformal prediction is able to produce substantially smaller prediction regions. Third, unlike the Bayesian learning framework, conformal predictors do not require any assumptions about the data priors.

1.2 Problem

Although conformal prediction constitutes a powerful framework intended to improve the capabilities of predictive machine learning algorithms, its relative nonubiquitousness (compared to traditional predictive methods) leads to questions regarding the proper design and use of a conformal prediction system. Specifically, to an analyst unfamiliar with the conformal prediction framework, it is not necessarily clear how a conformal prediction system should be properly configured in order to maintain desired properties, i.e., *validity* (a well-calibrated error probability) and *efficiency* (small prediction regions). Simultaneously, the way of interpreting the predictions output by a conformal predictor, which differs substantially from traditional classification and regression algorithms, may lead users unfamiliar with the framework to misconstrue the guarantees provided by the conformal prediction framework.

This thesis provides guidance for the proper design and interpretation of conformal prediction models. Specifically, three avenues of inquiry, detailed below, are considered: *Interpretability*; *Choosing a nonconformity function*; and *Creating conformal predictor ensembles*.

Interpretability: Conformal predictors are inherently based on p -value statistics, through the use of hypothesis testing. Although this leads to conformal predictors expressing powerful and statistically meaningful details about their predictions, appropriate use of and reliance on such predictions requires a correct understanding of p -value semantics. However, p -value statistics are often misinterpreted and reasoning that relies on p -value statistics can lead to incorrect conclusions about the underlying phenomena being scrutinized [Badenes-Ribera et al., 2015; Colquhoun, 2014, 2017; Daniel, 1998; Goodman, 2008; Johnson, 1999]. As conformal predictors are so closely tied to p -value statistics, and p -values seemingly so difficult to interpret, it appears only natural to seek out alternative means of presenting the predictions produced by a conformal prediction model. Thus far, however, the issue has not been addressed in conformal prediction literature.

Choosing a nonconformity function: A central design choice in creating any conformal predictor is that of selecting a suitable nonconformity function. In most circumstances, the nonconformity function will be based on a traditional predictive model (e.g., a classifier or a regression model), and it is natural to assume that more accurate underlying models lead to better conformal predictors. Although this assumption generally appears to hold, it is not clear how properties of the underlying model, other than predictive accuracy, affect the quality of the resulting conformal predictor.

Although several studies exist that assess conformal predictor performance using different types of underlying machine learning algorithms (e.g., k -nearest neighbors [Papadopoulos et al., 2011], random forests [Devetyarov and Nouretdinov, 2010], decision trees [Johansson et al., 2013a,b], artificial neural networks [Papadopoulos, 2008] and gradient boosting [Vovk, 2015]), little research has tied the performance of conformal predictors to more fundamental properties of the underlying algorithms (e.g., learner stability).

Creating conformal predictor ensembles: It is well-known in the machine learning community that combining multiple individual models into a single aggregate model (i.e., an *ensemble*) often is a good strategy for obtaining improved prediction quality. Typically, such ensemble strategies involve aggregating, in some way or another, the predictions made by the ensemble members to produce a single combined output. As conformal predictor outputs are based on p -value statistics, combining their predictions is subject to the common concerns regarding p -value aggregation (e.g., independence or interdependence between the multiple p -values), and it is not immediately obvious how ensemble learning theory can be leveraged in order to improve the predictive performance of conformal predictors.

The primary strategy for conformal prediction ensemble construction, *cross-conformal prediction*, often appears to perform well empirically, but the method is known to not produce valid conformal predictors in general [Vovk, 2015], a limitation that also applies to variants of the strategy, including *aggregated conformal predictors* [Carlsson et al., 2014] and *bootstrap conformal predictors* [Vovk, 2015]. Thus, seeking out alternative strategies for conformal predictor ensemble construction, ideally strategies that produce statistically valid conformal predictors, appears necessary.

1.3 Research Questions

How can we create conformal predictors that are simultaneously efficient, valid and interpretable?

- *How can we provide an interpretation of conformal predictors that does not rely on an understanding of p -value statistics?* (Papers I, III, V)
- *How does the choice of nonconformity function affect the efficiency, validity, and interpretability of a conformal predictor?* (Papers I, II, IV, VI)
- *How can informationally efficient and statistically valid ensembles be created from conformal predictors?* (Papers IV, VI)

1.4 Contributions

- **Alternative interpretations of conformal predictors.**

In Papers III and V, we propose two different methods for post-processing the predictions made by a conformal classification model, in such a way that the details of the underlying p -value statistics are effectively hidden from the end-user. The proposed methods instead present predictions on alternative forms: in Paper III, we provide a method for re-calibrating the conformal predictor's p -values to more closely resemble ordinary probabilities; and, in Paper V we provide a method for transforming a conformal classification model into an ordinary classifier (i.e., one that only outputs a single class label per test object) with the quality that the expected error frequency on the test set (or an ordered subset thereof) can be estimated without revealing the true labels.

In Paper I, we address a particular detail in the standard procedure for conformal regression that can potentially result in erroneous conclusions. That is, as the prediction intervals produced are not balanced, the probability of a prediction error falling above or below the prediction interval is unknown. We propose an alternative method for finding the prediction interval boundaries that ensures the resulting intervals are automatically balanced.

- **An understanding of the relationship between efficiency, validity and properties of the nonconformity measure.**

In Paper II, we systematically examine the relationship between conformal predictor performance and classifier instability, i.e., the tendency

of some learning algorithms to produce substantially different learned models given only small changes in training data. In addition, we provide guidance on the suitability of certain types of nonconformity functions given particular conformal prediction strategies. Specifically, we show that unstable learning algorithms are not well-suited for use in the transductive conformal prediction procedure.

Papers IV and VI further study the relationship between classifier instability and conformal predictor performance, but in the context of conformal predictor ensemble construction. We provide strong evidence that unstable learning algorithms tend to negatively affect efficiency and calibration in conformal prediction ensembles.

Paper I additionally provides insights into how the choice of nonconformity function affects conformal predictor performance, as the core of our alternative conformal regression procedure is a novel nonconformity measure.

- **Insights into conformal prediction ensemble design.**

In Paper IV, we provide a thorough analysis of state-of-the-art conformal prediction ensemble strategies, and illustrate conditions under which they produce unsuitable models. Specifically, we show that the introduction of excessive randomness (e.g., by the use of unstable underlying algorithms) tends to produce inefficient and poorly calibrated conformal predictor ensembles.

In Paper VI, we provide an alternative strategy for training conformal prediction ensembles that addresses some of the issues pertaining to state-of-the-art methods illustrated in Paper IV. Although we are unable to show that the proposed strategy is automatically well-calibrated, we show that the method should perform empirically better than existing alternatives.

1.5 Delimitations

While the conformal prediction framework is applicable in many contexts, including anomaly detection [Laxhammar and Falkman, 2010], clustering [Cherubin et al., 2015] and recommender systems [Kagita et al., 2017], this thesis exclusively addresses applications of conformal prediction to classification and regression problems, i.e., prediction problems where the response variable is either a (single) categorical variable or a real-valued variable.

For conformal classifiers, empirical investigations have been exclusively performed on binary classification problems. With the exception of Paper III, this choice was made primarily to more effectively communicate our research. We expect the results presented in Papers II, IV, V and VI to be applicable also to multi-class problems, but we are limited to drawing conclusions about binary classification problems due to our empirical strategy. The procedure presented in Paper III was specifically designed for binary classification problems, and the strategy as presented is not applicable to multi-class problems.

1.6 Disposition

The remainder of this thesis is organized as follows. Chapter 2 provides an overview of conformal prediction and related theory relevant to this thesis. In chapter 3, we outline the methodology followed in this thesis, including a description of research strategies, data sets, and algorithms used. Chapter 4 summarizes the results found throughout the thesis work, and Chapter 5 provides concluding remarks where we answer our research questions and propose directions for future research.

2. Conformal Prediction

You know what you are? You're a beard with an idiot hanging off it.

—Bernard Black

2.1 Theoretical Foundations

Conformal predictors (CPs) are a family of predictive models that associate each of their predictions with a measure of statistically valid confidence [Vovk et al., 2005]. We assume that there is some *example space* or *pattern space*¹ $\mathbf{Z} := \mathbf{X} \times \mathbf{Y}$, where $\mathbf{X}$ is a non-empty *object space* and $\mathbf{Y}$ is a *label space* consisting of at least two distinct elements. Given an object $\vec{x} \in \mathbf{X}$, the goal of a conformal predictor is then to output a *prediction region* $\Gamma \subseteq \mathbf{Y}$ such that the probability $P(y \in \Gamma)$ is known *a priori*.

To accomplish our task, we further assume that we are given a sequence of historical examples

$$\zeta^k = (\vec{x}_1, y_1), \dots, (\vec{x}_k, y_k), \quad (2.1)$$

where both objects and labels are known. For brevity, we will use the definition $z_i = (\vec{x}_i, y_i)$ when appropriate. We additionally require that the sequence

$$\zeta^{k+1} = (\vec{x}_1, y_1), \dots, (\vec{x}_k, y_k), (\vec{x}_{k+1}, y_{k+1}), \quad (2.2)$$

where z_{k+1} is our *test pattern* (i.e., the object $\vec{x}$ for which we wish to predict y), is *exchangeable*.

Definition 2.1.1 (Exchangeability) *An ordered sequence $\zeta^{k+1} = z_1, \dots, z_k$ is exchangeable if $P(z_1, \dots, z_k) = P(z_{\pi(1)}, \dots, z_{\pi(k)})$, where $\pi(1), \dots, \pi(k)$ is any permutation of the indices $1, \dots, k$.*

In practice, our assumption of exchangeability is weaker than that of the standard *independent and identically distributed (i.i.d.)* assumption, as we require

¹We will refer to test instances $z \in \mathbf{Z}$ (where the true value of y is either known or unknown) as *patterns*, and historical instances $z \in \mathbf{Z}$ (where the true value of y is always known) as either *patterns* or *examples*, interchangeably.

the examples to be identically distributed, but not necessarily independent (a random sample drawn without replacement, e.g., is exchangeable but not strictly *i.i.d.*). Since we assume that ζ^{k+1} is exchangeable, the order in which observations are obtained is uninteresting in general. Because we are only concerned with knowing the position of z_{k+1} , we can relax the definition of ζ^{k+1} to be a *bag* (a multi-set, where we allow duplicate entries)

$$\zeta^{k+1} = \{\vec{x}_1, y_1\}, \dots, \{\vec{x}_k, y_k\} \cup \{\vec{x}_{k+1}, y_{k+1}\}. \quad (2.3)$$

For simplicity, we will commonly refer to ζ^* simply as sets.

Given an exchangeable set ζ^{k+1} , the task is to predict the value y_{k+1} based on the information contained in ζ^k and $\vec{x}_{k+1}$. In general, this involves learning a function $g : \mathbf{X} \rightarrow \Gamma$, where Γ is some approximation of $\mathbf{Y}$, and applying the learned rule g to $\vec{x}_{k+1}$ to obtain a prediction $\hat{y}_{k+1} = g(\vec{x}_{k+1})$ that is, hopefully, approximately equal to y_{k+1} . Traditionally, we expect that $\Gamma \cong \mathbf{Y}$, i.e., the prediction rule g maps $\vec{x}_{k+1} \in \mathbf{X}$ onto any one value $\hat{y}_{k+1} \approx y_{k+1} \in \mathbf{Y}$. In conformal prediction however, we let $\Gamma = 2^{\mathbf{Y}}$, where $2^{\mathbf{Y}}$ is the power set of $\mathbf{Y}$, i.e., the output of a conformal predictor is (possibly) multi-valued. We then expect that $y_{k+1} \in (\Gamma_{k+1} \in 2^{\mathbf{Y}})$ with some known probability. Specifically, a conformal predictor will output a prediction region Γ_{k+1}^ϵ such that $P(y_{k+1} \notin \Gamma_{k+1}^\epsilon) \leq \epsilon$, where $1 - \epsilon$ is the *confidence* of the prediction, which is user-specified. Hence, conformal predictors are exactly valid *confidence predictors* [Vovk et al., 2005].

Definition 2.1.2 (Validity) A confidence predictor is valid iff $P(y_{k+1} \in \Gamma_{k+1}^\epsilon) \geq 1 - \epsilon$.

To construct the prediction rule g , a conformal predictor leverages a first-order function $f : \mathbf{Z}^* \times \mathbf{Z} \rightarrow \mathbb{R}$, normally referred to as a *nonconformity measure*, which scores an object $z \in \mathbf{Z}$ according to how well it fits into the empirical distribution described by the set $\zeta^{k+1} \subseteq \mathbf{Z}^*$. We define the *nonconformity score* of an object $z_i \in \zeta^{k+1}$ as

$$\alpha_i = f[\zeta^{k+1}, z_i], \quad (2.4)$$

where f may be any function that maps $\mathbf{Z}^* \times \mathbf{Z}$ onto $\mathbb{R}$ without affecting validity as long as ζ^{k+1} is exchangeable.¹

We can let $\alpha_1, \dots, \alpha_k$ be the nonconformity scores of a set of examples with known labels, i.e., an empirical estimation of $A : f[\mathbf{Z}^*, \mathbf{Z}]$. Assume that $\alpha_n \leq \alpha_m$ for $n, m \in 1, \dots, k$. Let Q_i denote the i -quantile of A , such that $P[A \leq$

¹We will later put more extensive expectations on f in order to retain other useful qualities, in addition to validity.

$Q_i(A)] \leq i$ where $i \in [0, 1]$, and let $\rho(t) = P[A \leq \alpha_t]$ for $t \in 1, \dots, k$. Given an ordered set $\alpha_1, \dots, \alpha_k$, it is trivial that

$$\rho(\lfloor ik \rfloor) \leq P[A \leq Q_i(A)] \leq \rho(\lceil ik \rceil), \quad (2.5)$$

i.e.,

$$\lim_{k \rightarrow \infty} Q_i = \alpha_{\text{round}(ik)}. \quad (2.6)$$

Assume now that we have a set ζ^{k+1} containing k training examples with nonconformity scores $\alpha_1, \dots, \alpha_k$, together with a single test pattern with a known true label y_{k+1} and nonconformity score α_{k+1} . Permute the indices of $\alpha_1, \dots, \alpha_{k+1}$, as $\alpha_{\pi(1)}, \dots, \alpha_{\pi(k+1)}$, where $\pi(i) \in 1, \dots, k+1$, such that $\pi(n) < \pi(m) \rightarrow \alpha_n \leq \alpha_m$. Given our assumption of exchangeability, we can produce a statistic on the values $\alpha_1, \dots, \alpha_{k+1}$ that possesses an interesting, predictable quality—that all values $\pi(k+1) \in 1 \dots k+1$ are unconditionally equiprobable:

$$\forall i \in 1 \dots k+1 : P[\pi(k+1) = i] = \frac{1}{k+1}. \quad (2.7)$$

From Equation (2.6), we may then obtain

$$\lim_{k \rightarrow \infty} Q_{\frac{\pi(k+1)}{k+1}} = \alpha_{\frac{\pi(k+1)}{k+1} k+1} = \alpha_{\pi(k+1)}, \quad (2.8)$$

hence,

$$P[A \leq \alpha_{k+1}] = \frac{\pi(k+1)}{k+1}. \quad (2.9)$$

Since z_{k+1} is our test pattern, we normally do not have knowledge of y_{k+1} (if we do, making a prediction is a rather uninteresting task). To make predictions, instead we make assumptions about the nature of y_{k+1} , by postulating a hypothetical label $\tilde{y} \in \mathbf{Y}$ and tentatively assigning it as the label for $\vec{x}_{k+1}$, forming the pattern $(\vec{x}_{k+1}, \tilde{y})$. If $\tilde{y} = y_{k+1}$ (and ζ^{k+1} is exchangeable), then $\alpha_{k+1}^{\tilde{y}}$ becomes predictable in nature—specifically, if $\tilde{y} = y_{k+1}$, then the property given in Equation 2.7 will hold, but if $\tilde{y} \neq y_{k+1}$, it may not. As we will see in the following sections, this property allows us to use Equation 2.9 to devise a simple statistical test, that can tell us whether $\tilde{y}$ is likely to be the correct label for $\vec{x}_{k+1}$.

2.2 Nonconformity Functions

We allow f to be any function that maps $\mathbf{Z} \times \mathbf{Z}^*$ onto $\mathbb{R}$, without affecting validity (we only require that ζ^* is exchangeable). However, in order to provide

informationally efficient predictions (i.e., small regions $\Gamma \subseteq \mathbf{Y}$), we expect that $\alpha_i \propto P(z_i \notin \mathbf{Z}^*)$. That is, we expect f assigns small nonconformity scores to common patterns, and larger nonconformity scores to uncommon patterns. To retain validity and maximize efficiency, we generally pose the following two requirements on the nonconformity function:

I The value of $\alpha_{k+1}^{y_{k+1}}$ does not matter in general for either validity or efficiency. We care only that it follows the same distribution as $\alpha_1, \dots, \alpha_k$. That is, the set ζ^{k+1} must be exchangeable, and all elements $z_i \in \zeta^{k+1}$ must be treated equally.

II We wish the value of $\alpha_{k+1}^{\tilde{y}}$ to be as large as possible, whenever $\tilde{y} \neq y_{k+1}$.

Thanks to decades of machine learning research, a multitude of candidate functions exist in the many algorithms that generate function approximations in the form of predictive rules (i.e., classification and regression algorithms). We can easily construct a general nonconformity measure on the form

$$f[\zeta^*, z_i] = \Delta[h(\vec{x}_i), y_i], \quad (2.10)$$

where h is a predictive model induced from the set ζ^* .

From any well-constructed predictive model, we should expect both I and II to hold. Criterion II appears intuitively automatic for an accurate predictive model: if $(\vec{x}_{k+1}, \tilde{y})$ is an unlikely pattern (e.g., $\tilde{y}$ is not the correct class y_{k+1}), then an accurate model h should in general make a larger error on $(\vec{x}_{k+1}, \tilde{y})$ than it does on $(\vec{x}_{k+1}, y_{k+1})$. Criterion I hinges on the assumption of exchangeability as well as the requirement that we treat all patterns $z_1, \dots, z_{k+1}$ equally (i.e., the training set of h must include either all of $z_1, \dots, z_{k+1}$, or none of them). The following sections describe the process of constructing a conformal predictor using a nonconformity function based on an arbitrary predictive learning algorithm.

2.3 Classification by Transduction

In a transductive conformal predictor [Gammerman et al., 1998; Saunders et al., 1999; Vovk et al., 2005], we solve criterion I by including z_{k+1} in the training data. That is, we induce h from the set ζ^{k+1} . Since we do not strictly require that the nonconformity function uses a learned predictive function, we use the more general notation $\alpha_i = f[\zeta^*, z_i]$; however, in most practical circumstances, we may assume (without loss of generality) that f is a function induced from ζ^* using a learning algorithm H that produces a predictive model h , coupled with some error measure $\Delta[\hat{y}_i, y_i]$.

Given a training set ζ^k and a test object $\vec{x}_{k+1}$, we tentatively assign the label $\tilde{y} \in \mathbf{Y}$ to $\vec{x}_{k+1}$, and construct the extended training set $\zeta^{k+1} = \zeta^k \cup \{(\vec{x}_{k+1}, \tilde{y})\}$. Nonconformity scores are then computed for each $z_i \in \zeta^{k+1}$ as either

$$\alpha_i^{\tilde{y}} = f \left[\zeta^{k+1}, z_i \right] \quad (2.11)$$

or

$$\alpha_i^{\tilde{y}} = f \left[\zeta^{k+1} \setminus z_i, z_i \right]. \quad (2.12)$$

Equation 2.12 is a *leave-one-out* transductive conformal predictor, where the nonconformity score of each pattern $z_i \in \zeta^{k+1}$ is computed from the remainder of the set. Whether we choose to compute the nonconformity scores using Equation 2.11 or 2.12 generally will not affect validity of the resulting conformal predictor, but it might affect its efficiency.¹

Following Equation 2.9, we know that

$$P \left[A \leq \alpha_{k+1}^{y_{k+1}} \right] = \frac{\pi(k+1)}{k+1}. \quad (2.13)$$

That is, if $\tilde{y} = y_{k+1}$, then we should expect that the rank of nonconformity score of the test pattern be uniformly distributed as $\pi(k+1) \sim U[1, k+1]$, giving $\frac{\pi(k+1)}{k+1} \sim U[0, 1]$. On the other hand, if $\tilde{y} \neq y_{k+1}$, and f fulfills criterion II given in Section 2.2, we should expect, on average, that

$$P \left[A \leq \alpha_{k+1}^{y_{k+1}} \right] \leq P \left[A \leq \alpha_{k+1}^{\tilde{y}} \right], \quad (2.14)$$

which suggests the following: if we assume a uniform distribution of test instance nonconformity ranks and reject predictions following a one-tailed test with a significance level $\varepsilon \in (0, 1)$, then we should expect true labels to be rejected with a frequency of ε and false labels to be rejected with a frequency greater than ε .

Hence, once nonconformity scores are computed for the training examples and the tentatively labeled test pattern, a p -value statistic (assuming a uniform distribution) is calculated as

$$p_{k+1}^{\tilde{y}} = \frac{\left| \left\{ z_i \in \zeta^{k+1} : \alpha_i^{\tilde{y}} \geq \alpha_{k+1}^{\tilde{y}} \right\} \right|}{|\zeta^{k+1}|}. \quad (2.15)$$

These p -values are calculated for each possible label $\tilde{y} \in \mathbf{Y}$, each time recomputing $\alpha_1^{\tilde{y}}, \dots, \alpha_{k+1}^{\tilde{y}}$ from $z_1, \dots, z_k, (\vec{x}_{k+1}, \tilde{y})$, and the final prediction region is constructed as

$$\Gamma_{k+1}^\varepsilon = \left\{ \tilde{y} \in \mathbf{Y} : p_{k+1}^{\tilde{y}} > \varepsilon \right\}, \quad (2.16)$$

¹Paper II deals with this issue in particular.

where ε is any user-specified significance level in $(0, 1)$.

The above process describes a conservatively valid conformal predictor, where $P(y_{k+1} \notin \Gamma_{k+1}^\varepsilon) \leq \varepsilon$. This conservative validity is a result of the possibility of ties occurring, where the nonconformity score obtained for the test object is identical to the nonconformity scores of one or more of the training examples. An exactly valid (*smoothed*) p -value can be computed by introducing a randomized term that breaks such ties uniformly, as

$$p_{k+1}^{\bar{y}} = \frac{\left| \left\{ z_i \in \zeta^{k+1} : \alpha_i^{\bar{y}} > \alpha_{k+1}^{\bar{y}} \right\} \right| + \theta_{k+1}^{\bar{y}} \left| \left\{ z_i \in \zeta^{k+1} : \alpha_i^{\bar{y}} = \alpha_{k+1}^{\bar{y}} \right\} \right|}{|\zeta^{k+1}|}, \quad (2.17)$$

where $\theta_{k+1}^{\bar{y}} \sim U[0, 1]$.

The computational time complexity of making a single prediction using a leave-one-out transductive conformal predictor is $O[(k+1) \times |\mathbf{Y}| \times (C_t(k) + C_p(1))]$, where $C_t(k)$ is the cost of fitting h using k training examples, and $C_p(1)$ is the cost of making a single prediction using h . The costs of performing the steps in Equations 2.15 and 2.16 are omitted, as they are normally inconsequential in comparison to $C_t + C_p$.

The analogous time complexity for making a single prediction using a transductive conformal predictor that does not operate in the leave-one-out fashion is smaller, but still prohibitive (for long test sequences): $O[|\mathbf{Y}| \times (C_t(k+1) + C_p(k+1))]$, where $C_t(k+1)$ is the cost of fitting h using $k+1$ training examples, and $C_p(k+1)$ is the cost of making $k+1$ predictions using h .

2.4 Classification by Induction

In an inductive conformal predictor [Papadopoulos et al., 2002; Vovk et al., 2005], training the model and making predictions for test objects are two distinct procedures, as with most common predictive machine learning methods. Here, the training set ζ^k is divided into two non-empty disjoint subsets: a *proper training set* ζ^t and a *calibration set* ζ^c .

Nonconformity scores are computed for the examples in the calibration set, $z_i \in \zeta^c$, as

$$\alpha_i = f[\zeta^t, z_i], \quad (2.18)$$

and a nonconformity score is computed for the tentatively labeled test pattern as

$$\alpha_{k+1}^{\bar{y}} = f[\zeta^t, (\vec{x}_{k+1}, \bar{y})]. \quad (2.19)$$

The key distinction between the transductive and inductive modes of conformal classification relates to criterion I given in Section 2.2: to ensure that $\alpha_1, \dots, \alpha_k, \alpha_{k+1}^{y_{k+1}}$ are exchangeable, we must treat the corresponding objects equally. In the field of machine learning, it is well known that if some objects are used to induce a model and others are not, we cannot generally consider them as treated equally. The transductive conformal predictor circumvents this problem by including all instances—even the test object—in all steps of the predictive process, including training the underlying model and performing statistical inference, which leads to a large computational cost since the underlying model must be recomputed for each new test object. In an inductive conformal predictor, only the proper training is used to train the underlying model, whereas the calibration set is used for statistical inference. This decoupling of training and inference means that we only need to (implicitly) include the test object in the calibration set and not the proper training set, allowing us to train the underlying model only a single time and reuse it for multiple predictions.

To obtain a p -value from an inductive conformal predictor, we compare the nonconformity score of the tentatively labeled test object only to the nonconformity scores of the calibration set, as

$$p_{k+1}^{\tilde{y}} = \frac{\left| \left\{ z_i \in \zeta^c : \alpha_i > \alpha_{k+1}^{\tilde{y}} \right\} \right| + \theta_{k+1}^{\tilde{y}} \left(\left| \left\{ z_i \in \zeta^c : \alpha_i = \alpha_{k+1}^{\tilde{y}} \right\} \right| + 1 \right)}{|\zeta^c| + 1}, \quad (2.20)$$

where $\theta_{k+1}^{\tilde{y}} \sim U[0, 1]$.

As before, a prediction region is constructed as

$$\Gamma_{k+1}^\varepsilon = \left\{ \tilde{y} \in \mathbf{Y} : p_{k+1}^{\tilde{y}} > \varepsilon \right\}. \quad (2.21)$$

The computational time complexity of training an inductive conformal predictor is $O[C_t(k - q) + C_p(q)]$, where $C_t(k - q)$ is the cost of training the underlying model using $k - q$ training instances (the size of the proper training set) and $C_p(q)$ is the cost of making q predictions (the size of the calibration set).

The cost of making a single prediction is simply $O[C_p(1)]$, again disregarding the cost of computing Equations 2.20 and 2.21. Depending on the qualities of the underlying model, these two equations, with time complexities $O[\log_2 n]$ (assuming the list of calibration scores is sorted) and $O[|\mathbf{Y}|]$, respectively, might either dominate or be dominated by the cost of $O[C_p(1)]$.

2.5 Regression by Induction

While theoretically possible, constructing a transductive conformal predictor for regression is difficult in general (given that we need to retrain the underlying model once for each possible output value of the test object). Therefore, for regression problems, inductive conformal predictor variants are generally preferred.

To construct an inductive conformal regression model [Papadopoulos, 2008; Papadopoulos et al., 2002; Vovk et al., 2005], we begin, as before, by dividing the original training data into a proper training set and a calibration set, and defining a nonconformity function as

$$\alpha_i = f[\zeta^t, z_i]. \quad (2.22)$$

Although a conformal classifier can effectively be constructed using an arbitrary nonconformity function, a conformal regressor requires that f is (partially) invertible for efficient computation. In addition, unlike the case with a conformal classifier, the formulation of the predictive step of a conformal regressor depends on the partial inverse of f ; hence, we assume the use of the common (partially invertible) nonconformity function

$$\alpha_i = |y_i - h(\vec{x}_i)|, \quad (2.23)$$

where $h(\vec{x}_i)$ is the prediction made by a regression model trained on ζ^t .

Assuming that Equation 2.23 was used as the nonconformity function, and that the calibration scores are sorted in descending order, we then construct a prediction region following

$$\Gamma_{k+1}^\varepsilon = h(\vec{x}_{k+1}) \pm \alpha_\delta, \quad (2.24)$$

where $\delta = \lfloor \varepsilon(1 + |\zeta^c|) \rfloor$, i.e., the index of the ε -percentile calibration score (rounded down).

Since α_δ is added/subtracted from the bare prediction of the underlying model to construct the upper/lower bounds of the prediction region in Equation 2.24, it should be clear that all prediction regions for a full test set will necessarily have the same width (for a given ε). This behavior is often unwanted, since we want the width of the prediction interval to reflect the certainty of the prediction.

To obtain prediction intervals whose widths vary with prediction certainty, we modify the nonconformity function as

$$\alpha_i = \frac{|y_i - h(\vec{x}_i)|}{\sigma_i + \beta}, \quad (2.25)$$

where σ_i is the estimated difficulty of predicting the output of $\vec{x}_i$, given, e.g., by a separate regression model trained on the residual errors of h , and β is a hyper-parameter regulating the influence of σ_i .

If Equation 2.25 is used as the nonconformity function, we construct the prediction region as

$$\Gamma_{k+1}^\varepsilon = h(\vec{x}_{k+1}) \pm (\sigma_{k+1} + \beta)\alpha_\delta, \quad (2.26)$$

where, again, $\delta = \lfloor \varepsilon (1 + |\zeta^c|) \rfloor$ and σ_{k+1} is the estimated difficulty of the test object.

2.6 Aggregated Conformal Prediction

When comparing transductive and inductive conformal predictors, we see a trade-off with regards to informational efficiency and computational efficiency: although a transductive conformal predictor can take advantage of the entire training set both for training and calibration and therefore maximizing the expected informational efficiency, their computational complexity is prohibitively expensive for any substantial data set size. An inductive conformal predictor, on the other hand, is computationally much more efficient; however, as only parts of the data can be used for training and calibration respectively, the expected informational efficiency decreases.

Aggregated conformal predictors provide a middle-ground, borrowing from the concept of ensemble learning. Here, instead of constructing a single underlying model that is calibrated on a single calibration set, a series of models are constructed using multiple subsets of the initial training data, and calibration is performed, per ensemble member, using those training examples not included in the training data subset.

2.6.1 Cross-Conformal Predictors

Aggregated conformal predictors were originally introduced by Vovk [2015] in the form of cross-conformal predictors, which are a type of conformal predictor ensemble based on the cross-folding scheme typically seen in cross-fold validation. Here, the original training data ζ is divided into a number of (approximately) equally sized disjoint subsets, $\zeta_1^c, \dots, \zeta_n^c$, called *folds* (we use the c exponent to denote the fact that these folds are used much in the same manner as the calibration sets in an inductive conformal predictor). For each fold, we take the complement $\zeta_m^t = \zeta \setminus \zeta_m^c$ to be a training set from which we induce a predictive model h_m . We define multiple nonconformity functions, one per fold, as

$$\alpha_{i,m} = f_m [\zeta_m^t, (\vec{x}_i, y_i)], \quad (2.27)$$

and apply each nonconformity function f_m only to the calibration (sub-)set ζ_m^c . This gives us a single nonconformity score for each of the instances in ζ , where the scores from each group $\zeta_m^c \subset \zeta$ are obtained using a separate underlying model.

To make a prediction for a test object, we calculate the p -values for the possible labels as

$$p_{k+1}^{\tilde{y}} = \frac{\sum_{m=1}^n \left| \left\{ z_{i,m} \in \zeta_m^c : \alpha_{i,m} \geq \alpha_{k+1,m}^{\tilde{y}} \right\} \right| + 1}{|\zeta| + 1}, \quad (2.28)$$

or using a smoothed tie-breaking variant. We construct the prediction region as before, rejecting any label $\tilde{y}$ for which $p_{k+1}^{\tilde{y}} < \varepsilon$.

2.6.2 Bootstrap Conformal Predictors

Bootstrap conformal predictors, also described by Vovk [2015], are closely related to cross-conformal predictors, and are constructed in a similar manner. Here, however, instead of folding the dataset to obtain disjoint calibration sets, we draw multiple bootstrapped training sets $\zeta_1^t, \dots, \zeta_n^t$ and take the calibration sets to be their complements $\zeta_m^c = \zeta \setminus \zeta_m^t$. As before, we train an underlying model and construct a nonconformity score analogous to Equation 2.27 for each of the training sets. Each nonconformity function is then applied to its corresponding calibration set.

The p -values of a bootstrap conformal predictor are calculated as

$$p_{k+1}^{\tilde{y}} = \frac{\sum_{m=1}^n \left| \left\{ z_{i,m} \in \zeta_m^c : \alpha_{i,m} \geq \alpha_{k+1,m}^{\tilde{y}} \right\} \right| + t/|\zeta|}{|\zeta| + t/|\zeta|}, \quad (2.29)$$

where $t = \sum_{m=1}^n |\zeta_m^c|$ (i.e., the total size of the calibration sets).

2.6.3 Generalized Aggregated Conformal Predictors

While the definitions of cross-conformal predictors and bootstrap conformal predictors are inherently tied to their sampling strategies (cross-folding and bootstrap sampling, respectively), Carlsson et al. [2014] provide a generalized framework for aggregated conformal predictors that potentially allows for any sampling strategy of the training and calibration sets (under some conditions relating to exchangeability).

Here, we simply assume that the sets $\zeta_1^t, \dots, \zeta_n^t$ and $\zeta_1^c, \dots, \zeta_n^c$ are obtained using any sampling strategy such that $\forall \zeta_*^c : \zeta_*^c$ is exchangeable with ζ .

The nonconformity functions are defined, again, as

$$\alpha_{i,m} = f_m [\zeta_m^t, (\vec{x}_i, y_i)] . \quad (2.30)$$

The p -value computation then becomes a two-step procedure, where we first obtain a single p -value per fold (per test object and label) as

$$p_{k+1,m}^{\tilde{y}} = \frac{\left| \left\{ z_{i,m} \in Z_m^c : \alpha_{i,m} \geq \alpha_{k+1,m}^{\tilde{y}} \right\} \right| + 1}{|Z_m^c| + 1} . \quad (2.31)$$

To form the final p -value used during prediction, these fold- p -values are then aggregated simply by taking their mean:

$$p_{k+1}^{\tilde{y}} = \frac{1}{n} \sum_{m=1}^n p_{k+1,m}^{\tilde{y}} . \quad (2.32)$$

2.7 Mondrian Conformal Predictors

As we have noted in passing, without further qualification, the validity of a conformal predictor is *unconditional*. That is, before we know anything about our test object (its feature values or its label) or the particular circumstances that went into constructing our conformal predictor (the examples included in the training set, randomization effects of the learning algorithm used to construct the nonconformity function, etc.), we may say that the likelihood of making an erroneous prediction is exactly ε . Although such unconditional probabilities are generally easier to deal with, they might cause us trouble in many circumstances. We might, for example, consider a conformal predictor used for medical diagnosis; we may say that the probability of rejecting the correct diagnosis is ε unconditionally, without considering, for example, whether the patient is male or female. Once we have learned the patient's gender, we may then ask: Is the probability of making an erroneous prediction still ε ? At best, the answer is that we do not know, as we only stated the error probability ε unconditionally. At worst, we might have reason to expect that the conditional error probability is very unlikely to be ε , or even close to it [Löfström et al., 2015].

Vovk [2012] provides a treatment of different types of conditional validity of conformal predictors, two of which are of particular interest: the object-conditional conformal predictors and label-conditional (or class-conditional) conformal predictors originally discussed by Vovk et al. [2005]. Generally, such *conditional* or *Mondrian* conformal predictors allow us to divide the problem space into sub-spaces, and guarantee an error rate ε for each sub-space.

Formally, we define a mapping function $K : \mathbf{Z} \rightarrow \kappa_1, \dots, \kappa_m$, where $\kappa_i = K(z_i)$ is the sub-space (or *category*) to which the object z_i belongs.

One simple example of an object-conditional mapping function, where we attempt to solve our hypothetical problem in medical diagnosis given above, is

$$K(z_i) = \begin{cases} 0 & \text{if gender}(\vec{x}_i) = \text{female} \\ 1 & \text{if gender}(\vec{x}_i) = \text{male}, \end{cases}$$

which will allow us to guarantee an error rate of ε regardless of whether $\vec{x}_i$ represents a female or a male patient.

Similarly, we may construct a label-conditional mapping function as

$$K(z_i) = \begin{cases} 0 & \text{if } y_i = \text{iris-versicolor} \\ 1 & \text{if } y_i = \text{iris-virginica} \\ 2 & \text{if } y_i = \text{iris-setosa}, \end{cases}$$

if our goal is to create a conformal predictor whose error rate does not depend on the true label of the test object. As we can see in this example pertaining to the famous iris data set, we are not restricted to binary divisions, but may include any number of categories in our mapping function.

We might construct a mapping function that is conditional on both objects and labels, e.g., as

$$K(z_i) = \begin{cases} 0 & \text{if gender}(\vec{x}_i) = \text{female} \wedge y_i = \text{sick} \\ 1 & \text{if gender}(\vec{x}_i) = \text{female} \wedge y_i = \text{no-sick} \\ 2 & \text{if gender}(\vec{x}_i) = \text{male} \wedge y_i = \text{sick} \\ 3 & \text{if gender}(\vec{x}_i) = \text{male} \wedge y_i = \text{no-sick}, \end{cases}$$

effectively creating an orthogonal lattice (reminiscent of the art of Piet Mondrian) on top of the problem space, where validity is guaranteed separately in each cell of the lattice.

Given a particular mapping function, that provides a category for each training (or calibration) example, as well as test objects (with tentative labels), calculating a conditional p -value simply involves using the ordinary p -value function with the added condition that comparisons are only made between objects belonging to the same category, as

$$p_{k+1}^{\tilde{y}} = \frac{\left| \left\{ z_i \in \zeta^c : \alpha_i \geq \alpha_{k+1}^{\tilde{y}} \wedge K(z_i) = K(z_{k+1}) \right\} \right| + 1}{|\zeta^c| + 1}. \quad (2.33)$$

To preserve readability, we give only the non-smoothed variant of a conditional p -value, but be aware that a smoothed variant may easily be computed analogously to an ordinary inductive conformal predictor by splitting ties uniformly randomly.

3. Methodology

The only thing that's going to bring me inner peace is a beard-seeking missile!

—Bernard Black

3.1 Research Strategy

The scientific contributions presented in this thesis are grounded in a positivistic ontological view. As such, the results presented are obtained using two primary epistemological modes of inquiry: empiricism and deductivism. Primarily, our conclusions are underpinned by empirical results gathered through traditional experimental algorithm benchmarking, leveraging default methods for machine learning algorithm evaluation. Where we have deemed it appropriate, we provide mathematical reasons or proofs to support our claims.

3.2 Evaluation

In all papers, we have performed some type of empirical and experimental evaluation. These empirical assessments follow the general normative procedure of machine learning algorithm evaluation. This approach uses part of the available data to induce a predictive model and uses another disjoint part of the data to assess the induced model using relevant performance metrics.

3.2.1 Datasets

The empirical algorithmic evaluations rely on several standard-use benchmark datasets for classification algorithms (Table 3.1) and regression algorithms (Table 3.2). These are all obtained from the UCI [Asuncion and Newman, 2007], DELVE [Neal, 1998], and KEEL [Alcalá-Fdez et al., 2011] repositories. For classification sets, listed statistics are the total number of instances (#inst), the number of minority class examples (#min), the number of majority class examples (#max), and the percentage of minority class examples (%min). For

regression sets, the total number of instances (#inst) is listed together with the mean (μ) and standard deviation (σ) of the output variable.

Table 3.1: Binary classification datasets

Dataset	#inst	#min	#maj	%min	source	papers
balance-scale	576	288	288	50.0	UCI	II, III, V, VI
breast-cancer	286	85	201	29.7	UCI	II, III, V, VI
breast-w	699	241	458	34.5	UCI	II, III, V, VI
credit-a	690	307	383	44.5	UCI	II, III, V, VI
credit-g	1000	300	700	30.0	UCI	II, III, V, VI
diabetes	768	268	500	34.9	UCI	II, III, V, VI
haberman	306	81	225	26.5	UCI	II, III, V, VI
heart-c	303	138	165	45.5	UCI	II, III, V, VI
heart-h	294	106	188	36.1	UCI	II, III, V, VI
heart-s	270	120	150	44.4	UCI	II, III, V, VI
hepatitis	155	32	123	20.6	UCI	II, III, V, VI
ionosphere	351	126	225	35.9	UCI	II, III, V, VI
kr-vs-kp	3196	1527	1669	47.8	UCI	II, III, V, VI
labor	57	20	37	35.1	UCI	II, III, V, VI
liver-disorders	345	145	200	42.0	UCI	II, III, V, VI
mushroom	8124	3916	4208	48.2	UCI	II, III, V, VI
sick	3772	231	3541	6.1	UCI	II, III, V, VI
sonar	208	97	111	46.6	UCI	II, III, V, VI
spambase	4601	1813	2788	39.4	UCI	II, III, IV, V, VI
tic-tac-toe	958	332	626	34.7	UCI	II, III, V, VI

With the exception of Paper IV, the same collection of datasets was used throughout the studies so results of the papers could be compared. The particular datasets were chosen as a representative collection of typical benchmark problems, with varying characteristics in terms of number of rows, columns, class balance, and regression variable statistics. In Paper IV, the Spambase dataset was used specifically to produce results comparable to those presented by Vovk [2015] on cross-conformal predictors.

As our empirical assessments are primarily intended as a means to evaluate different conformal prediction strategies, given identical underlying models, no particular care was taken in preparing the datasets in order to maximize performance of the underlying learning algorithms. Hence, any missing values were replaced using simple column-average imputation, and no further feature engineering was performed on the datasets.

Table 3.2: Regression datasets

Dataset	#inst	μ	σ	source	papers
abalone	4177	0.24	0.14	DELVE	I
anacalt	4052	0.23	0.42	KEEL	I
bank8fh	8192	6.51	1.13	DELVE	I
bank8fm	8192	6.51	1.13	DELVE	I
bank8nh	8192	6.51	1.13	DELVE	I
bank8nm	8192	6.51	1.13	DELVE	I
boston	506	12.65	7.13	DELVE	I
comp	8192	1328125	421993	KEEL	I
concrete	1030	45.66	63.14	KEEL	I
cooling	768	2.81	1.55	KEEL	I
deltaA	7129	-0.0	0.0	KEEL	I
deltaE	9517	-0.0	0.01	KEEL	I
friedm	1200	0.49	0.28	KEEL	I
heating	768	2.81	1.55	UCI	I
istanbul	536	0.0	0.01	UCI	I
kin8fh	8192	0.0	0.18	DELVE	I
kin8fm	8192	-0.0	0.18	DELVE	I
kin8nh	8192	0.01	0.91	DELVE	I
kin8nm	8192	0.0	0.91	DELVE	I
laser	993	59.88	46.88	KEEL	I
mg	1385	0.93	0.23	UCI	I
mortage	1048	954.92	372.2	KEEL	I
plastic	1649	250.02	31.62	KEEL	I
puma8fh	8192	0.0	0.17	DELVE	I
puma8fm	8192	-0.0	0.17	DELVE	I
puma8nh	8192	0.0	0.35	DELVE	I
puma8nm	8192	-0.0	0.35	DELVE	I
quakes	2178	54.91	118.85	KEEL	I
stock	950	44.21	4.27	KEEL	I
treasury	1048	954.97	372.16	KEEL	I
wineR	1599	10.42	1.07	UCI	I
wineW	4898	10.51	1.23	UCI	I
wizmir	1461	34.28	2.44	KEEL	I

3.2.2 Learning Algorithms

While the validity of a conformal predictor is independent of the chosen non-conformity function, its efficiency is indeed strongly correlated with properties of the nonconformity function used. The nonconformity measures used are on the general form given in Equation 2.23 (i.e., a classification or regression model coupled with an appropriate error measure). The learning algorithms used in the included papers are the Random Forest learning algorithm [Breiman, 1996], classification and regression trees [Breiman et al., 1984], support vector machines [Cortes and Vapnik, 1995], and artificial neural networks.

In Paper I, which addresses regression problems, artificial neural networks were used as the underlying model, as previous research has found that neural networks are an appropriate basis for conformal regression models [Papadopoulos, 2008]. In the remaining papers, which address classification problems, the Random Forest algorithm was chosen as a representative for state-of-the-art classification algorithms, as it famously tends to have a good general performance for many types of classification problems [Liaw et al., 2002]. Decision trees and support vector machines were selected as representatives for unstable and stable classifier learning algorithms, respectively [Breiman, 1996; Buciu et al., 2006], providing a basis of comparison in Papers II, IV, and VI.

The implementations of these learning algorithms were supplied by the Scikit-learn library [Pedregosa et al., 2011] for the Python programming language. An implementation of the conformal prediction methods developed and evaluated throughout this thesis is accessible through GitHub¹. The decision to produce a new implementation of the conformal prediction framework was necessary due to no suitable implementation being publicly available. Since the results presented in this thesis rely on this implementation to a large degree, it was deemed appropriate to release the implementation publicly, to allow for review regarding potential errors or experimenter biases.

3.3 Criteria of Evaluation

Conformal predictors are subject to two fundamental desiderata: efficiency and validity. Validity refers to the property

$$P(y \notin \Gamma_{k+1}^\varepsilon) = \varepsilon,$$

i.e., the fact that the *a priori* probability of making an erroneous prediction is ε , unconditionally. This property is automatic for a conformal predictor, and

¹<https://github.com/donlnz/nonconformist>

does not in general depend on any property or assumption other than that of exchangeability.

Although validity is automatic, we may often wish to assess the empirical validity of a conformal predictor, i.e., the level of agreement between the theoretically expected error rate ϵ and the empirically observed error rate. In addition to serving as a “sanity check,” we expect that empirically assessing a conformal predictor’s validity reveals whether the error rate depends on additional factors, such as the chosen significance level or the true class label of the test object. Throughout this thesis, (empirical) validity is assessed simply by measuring the empirical error rate of a conformal predictor; for a perfectly calibrated predictor, we should expect that the empirical error rate is exactly ϵ .

In addition to establishing validity in our conformal predictors, we are often interested in assessing their efficiency, where efficiency denotes (in an abstract sense) the size of the prediction regions produced by a conformal predictor. We might evaluate the efficiency of a conformal predictor in a multitude of ways. Indeed, Vovk et al. [2016] detail several possible efficiency measures. Throughout this work, however, we will primarily refer to what Vovk et al. [2016] call the N -criterion (also called AvgC, for *average number of class labels* by Johansson et al. [2013a]) when discussing efficiency. Other measures, such as average p -value, provide fuller information about the overall efficiency of a conformal predictor as they provide an efficiency measure independent of ϵ , however, we consider these measures generally more difficult to interpret as they are arguably only meaningful to anyone already familiar with the conformal prediction framework. The N -criterion (or AvgC) gives us the average size of the prediction regions given a specific significance level, as

$$avgc = \frac{1}{n} \sum_{i=1}^n |\Gamma_{k+n}^{\epsilon}|,$$

where n is the size of the test set.

3.3.1 Evaluation Methods

To assess the performance of conformal predictors, with respect to metrics including efficiency and empirical validity, two common evaluation methods are used: repeated hold-out testing and k -fold cross-fold validation.

In hold-out testing, a randomly selected partition of the data is used to induce a model, and the remainder of the data, left out from training, is used to assess model performance. In repeated hold-out testing, the procedure is repeated multiple times (randomly selecting a new training set in each iteration). In

Paper I, repeated hold-out testing was performed, using 80% of the data for training and the remainder for testing on each dataset 100 times. An 80%/20% division was made to ensure an appropriate balance between training, calibration and testing data in each iteration.

In the k -fold cross-fold validation process, the dataset is split into k disjoint partitions, after which the data contained in $k - 1$ of the partitions is used to induce a predictive model whose predictive performance is assessed on the final partition, left out from training. The procedure is repeated k times, leaving out a new partition in each iteration, and statistical evaluation is finally performed, either on the performance results obtained from each individual fold or (as in our case) on an aggregation of the performance across all folds. Cross-fold validation was used in Papers II-VI.

We are aware that the use of both cross-validation and repeated hold-out testing presents an inconsistency between papers, and there is no immediate motivation for why both strategies were used. However, the two testing strategies serve the same purpose—to reduce variance in the obtained results. Moreover, we do not expect the choice of specific evaluation strategy to affect the results and conclusions presented in any of the papers.

4. Summary of Papers

I'm a quitter. I come from a long line of quitters. It's amazing I'm here at all.

—Bernard Black

The contributions of the papers included in this thesis are categorized according to three themes related to the research questions, outlined here: interpretability of predictions made by conformal predictors; properties of nonconformity functions; and the construction of conformal predictor ensembles. The results from each paper are summarized according to these three themes.

For the first theme, the interpretability of predictions made by conformal predictors, we address the potentially counter-intuitive nature of conformal predictors and p -values. Here, we provide a critique of conformal predictors and formulate some alternative interpretations of conformal predictors, in order to provide users of conformal predictors with an interpretation that does not require a deep understanding of p -value statistics.

For the second theme, the properties of nonconformity functions, we examine how the choice of nonconformity function affects efficiency (and, in certain aspects, even validity) of the resulting conformal predictor. Specifically, we apply the idea of classifier *stability* [Breiman, 1996] to nonconformity functions and investigate its relationship to the p -values produced by conformal predictors.

For the third theme, the construction of conformal predictor ensembles, we provide guidance on ensemble construction on two different levels: insight into suitable properties of members of a conformal predictor ensemble and insights into suitable ensemble generation and aggregation strategies.

4.1 Interpreting Conformal Predictors

It should be noted that the interpretability of a given model (or, perhaps, algorithm) can be judged from several angles. In machine learning literature, we typically take model interpretability to refer to the degree to which it is pos-

sible to assess the model’s “reasoning”, i.e., an interpretable model allows us to understand why a particular prediction is made, whereas a non-interpretable model appears as a black box that supplies an output (a prediction) based on some input. Successful efforts have been made to design conformal predictors that possess this quality of interpretability [Johansson et al., 2013a,b, 2014, 2018, 2019]. Here, however, we offer a slightly different view of interpretability, originally presented in Paper III, specific to conformal predictors: as the predictions made by a conformal predictor differ from those made by a traditional classifier (which might output only the single most probable label, or an estimate of the *a posteriori* class probabilities), they might indeed appear unusual to a user primarily familiar with traditional machine learning methods. Therefore, we critique the interpretability of the predictions made by an ordinary conformal predictor (ignoring the level of interpretability of the underlying model or nonconformity function).

In summary, for any (arbitrary) future prediction a conformal predictor makes, the probability is $1 - \epsilon$ that the prediction will include the correct output given the input. This statement intends to highlight two subtleties inherent in conformal prediction: when we make a prediction, the error probability is ϵ if we:

1. do not know the input values of the test object, and
2. do not know what labels will be included in the prediction region.

This first detail may seem counter-intuitive at first, but it follows naturally from the fact that conformal predictor validity is unconditional by default. We may contrast this to any type of machine learning model: for any model, we expect certain patterns to be more difficult to predict than others. For example, the model might have only seen very few similar examples during training or the particular pattern we are trying to predict lies on or close to some class boundary. This is also true for a conformal predictor, where similar factors may affect the probability of making an erroneous prediction for a particular test object but will not, for example, affect the empirical error rate observed on a randomly sampled test set. Section 2.7 discusses this phenomenon in further detail.

The second detail appears similarly confusing, but again has an analogue in traditional machine learning methods in the familiar precision metric commonly used in evaluating classifiers:

$$precision = \frac{TP}{TP + FP},$$

where TP is the number of true positive predictions (i.e., predictions that were correctly identified as belonging to the positive class), and FP is the number of false positive predictions (i.e., negative-class objects incorrectly predicted as belonging to the positive class). So, just as we might say that our classifier has an unconditional accuracy of $a\%$, we might proclaim that our classifier has a class-conditional accuracy (i.e., precision) of $b\%$.

4.1.1 Classification

For a conformal classifier, *ex post facto* reasoning about a prediction becomes somewhat more intricate, as we—in an absurd way—may unintentionally invalidate the conformal predictor’s claim of ε -validity (or, at least, our interpretation of this claim) simply by observing a prediction. Consider the following cases:

Case 1: Given a binary classification problem (classes 1 and -1), we obtain the following predictions for two separate test objects ($\vec{x}_1$ and $\vec{x}_1$) at $\varepsilon = e$:

$$\begin{aligned}\Gamma_1^\varepsilon &= \{-1\} \\ \Gamma_2^\varepsilon &= \{1\}\end{aligned}$$

Case 2: Given the same binary classification problem, we first obtain the predictions Γ_1^ε and Γ_2^ε . Next, we obtain the predictions for 98 separate (new) test objects ($\vec{x}_3, \dots, \vec{x}_{100}$) at $\varepsilon = e$:

$$\forall i \in [3, 100] : \Gamma_i^\varepsilon = \{-1, 1\}$$

Given only the predictions obtained in the first case, Γ_1 and Γ_2 , we might very well be inclined to assign the labels -1 and 1 to $\vec{x}_1$ and $\vec{x}_2$, respectively, and be content that we have assigned the correct labels with confidence $1 - e$.

Upon obtaining the next set of predictions, $\Gamma_3, \dots, \Gamma_{100}$, however (our second case) we find ourselves doubting not only the (additional 98) predictions we just obtained but also the initial two prediction we made. The premise (or, promise) that led to the predictions Γ_1 through Γ_{100} was this: any future prediction will be correct with unconditional probability $1 - e$. Conversely, we may state this as follows: any future prediction made will be erroneous with probability e . Before observing the predictions, we have no immediate reason to believe that the prediction for one particular object is more likely to

be correct than the prediction for any other object; however, after observing the predictions Γ_1 through Γ_{100} , we are faced with the fact that Γ_3 through Γ_{100} cannot possibly be incorrect (since they include all possible answers). If we knew from the onset that errors would be made with a probability (or frequency) of e , are we still content with assigning labels -1 and 1 to $\bar{x}_1$ and $\bar{x}_2$, respectively, with confidence $1 - e$? Naturally, after observing the full set of observations, we are instead inclined to assume that the total probability mass of erroneous predictions should be distributed over the predictions Γ_1 and Γ_2 , and a posterior adjustment of our confidence leads us to conclude that Γ_1 and Γ_2 are likely erroneous with a probability greater than e .

Although this perceived elevation of error probability of certain prediction regions stems from a forced misinterpretation of the output of the conformal predictor, it closely relates to another source of possible error rate elevation. That is, in our binary classification example, the predictions $\Gamma_3, \dots, \Gamma_{100}$ are ultimately uninteresting, as they provide very little information (in all these cases, the conformal predictor essentially refuses to make a prediction). Conversely, the small prediction regions Γ_1 and Γ_2 appear, on a superficial level, to provide us with useful information (as the conformal predictor has successfully managed to reject possible hypothesis labels) but in reality our posterior assessment of the predictions leads us to the realization that these two particular predictions are likely misleading rather than informative. We are thus left with the following:

- The prediction regions containing few potential class label assignments appear more “interesting”.
- The prediction regions containing few potential class label assignments are more likely to have excluded the true label.

Defining a proper and usable interpretation of conformal predictors (in a classification scenario) appears troublesome if we are only considering the bare prediction regions output by the conformal predictor. In Papers III and V, we address this by applying posterior reasoning to the predictions output by conformal classifiers, to provide more accessible predictions. The approaches provided in both papers essentially operate on a higher level of abstraction compared to the conformal predictors themselves. Whereas conformal predictors are designed to reason about and provide guarantees for individual objects, our proposed methods are designed to reason about the collection of prediction sets output by a conformal predictor, and utilizes posterior information regarding, for example, the frequency of predictions containing only a single class label.

Re-Evaluating Error Probabilities

In Paper III, we introduce a method of producing a posterior adjustment of the confidence associated with specific prediction regions, dependent on their particular size. Specifically, we address binary classification problems, where we can note the following (assuming that the possible class labels are 1 and -1):

- An empty prediction $\Gamma^\varepsilon = \emptyset$ is always incorrect (i.e., it never contains the true class label) independently of ε .
- A complete prediction $\Gamma^\varepsilon = \{-1, 1\}$ is always correct (i.e., it always contains the true class label) independently of ε .
- A singleton prediction, containing only a single class label, Γ^ε is correct with some probability related to ε .

As the (*a priori*) probability of *any* prediction being incorrect is exactly ε , it becomes a relatively easy task to adjust the confidence of a singleton prediction given that we can estimate the probability of a prediction set containing zero, one, or two class labels. The adjusted confidence of a singleton prediction is given as

$$\hat{\varepsilon} = \frac{\varepsilon - P(e)}{P(s)},$$

where $P(e)$ and $P(s)$ are the probabilities of obtaining an empty prediction and a singleton prediction, respectively.

Since conformal predictors assume that the data are exchangeable (and, by extension, generated by a stationary process), we should expect the empirically observed frequency of prediction sizes to converge. Therefore, we estimate the probability of empty, singleton, and complete predictions from the data. In addition, as we do not need to observe the true class labels to assess the prediction set size, we do not need to withhold any additional data from the model, but can simply do this estimation directly from the test set.

Noting that we are still left with the inconvenient situation that the error probability of a prediction $\{1\}$ is not necessarily the same as that of a prediction $\{-1\}$, we propose a class-conditional version of this posterior confidence adjustment. Assuming that the conformal predictor used is a class-conditional Mondrian conformal classifier, we can express the confidence of a prediction containing only the label c_i as

$$\hat{\varepsilon}_{c_i} = \frac{\varepsilon P(c_j)}{P(s_i)}, \quad (4.1)$$

where $P(c_j)$ is the prior probability of an object belonging to class c_j , which we can estimate from the training set, or the calibration set.

The empirical results presented in Paper III show that the adjusted confidence measures represented by $\hat{\epsilon}$ and $\hat{\epsilon}_{c_i}$ both correspond well with the observed error frequency of singleton predictions in unconditional and class-conditional cases, respectively. The cost of producing these adjusted confidence measures is that multiple predictions must be made before any reasonable adjustment can be made (the more predictions we observe, the more precisely we can estimate $P(e)$, $P(s)$, and $P(s_i)$ from the observed data); however, the provided benefit is quite clear. If we are mainly interested in providing informative predictions (i.e., prediction regions that are both small and correct), then informing users specifically on the quality of singleton predictions alleviates many of the confusing aspects of conformal classification that potentially lead to elevated error frequencies due to misinterpretation (see Section 4.1.1).

Introducing a Reject Option

A common use case for classification algorithms is the task of filtering a collection of objects to retrieve a subset of objects that possess some particular interesting quality, often referred to as *information retrieval*. A motivating example for this type of procedure is the drug discovery process: out of all possible molecular structures, we wish to identify specifically those that display some favorable characteristic such as water solubility or activity towards some target illness. One possible procedure for leveraging classification algorithms in order to achieve this is to train a classification model to predict a label related to the desired property. For example we may define a binary classification problem with labels $\{True, False\}$ denoting whether a particular molecule is sufficiently water soluble. After a classification model is trained using data describing molecules with known water solubility, the model is applied to a set of molecules with unknown water solubility to find promising candidates for further testing (e.g. *in vitro*).

Of course, in this process, we may wish to replace a traditional classification algorithm with a conformal classifier in order to ensure that we can assess the quality of the predictions made as making poor predictions in this context is costly. Unnecessary *in vitro* screening is expensive, and missing out on potential promising candidates carries a large opportunity cost. However, as we have laid out previously in this section, we are possibly setting ourselves up for a trap by using a (plain) conformal classifier. If we were to consider only those molecules whose corresponding output region is $\Gamma^e = \{True\}$, then those particular prediction regions are likely incorrect with a frequency (potentially

much) greater than ε .

Although we could conceivably choose to compute the adjusted confidence measures $\hat{\varepsilon}$ or $\hat{\varepsilon}_{c_j}$ described in the previous sub-section, in Paper V we present a procedure more suitable for this task. Specifically, we leverage conformal classification in constructing classification models with a reject option, i.e., classifiers that can refuse making a prediction if assigning any specific label appears too uncertain.

The basis for this reject-option classifier is a conformal classifier that produces confidence-credibility predictions: given a classification problem with labels $\{c_1, \dots, c_n\}$ and a test object $\vec{x}_k$, the conformal classifier is used to compute a p -value for each class, $p_k^{c_1}, \dots, p_k^{c_n}$. From this set of p -values we can construct a prediction in the following manner:

- The assigned class label is $\hat{y}_k = c_i$ that corresponds with the largest p -value, $p_k^{c_i}$.
- The confidence γ_k of this prediction is $1 - p_k^{c_j}$, where $p_k^{c_j}$ is the second largest p -value. This corresponds to the significance level $\varepsilon = 1 - \gamma_s$ at which all labels but c_i are rejected.
- The credibility μ_k of this prediction is $p_k^{c_i}$. This credibility corresponds to the confidence level $\varepsilon = \mu_s$ at which all labels, including c_i , are rejected. If μ_k is small, the conformal classifier considers all potential class labels as unsuitable for the test object.

Assuming that we have a set of test objects $\vec{x}_1, \dots, \vec{x}_m$, we can obtain predictions on the form defined above as $(\hat{y}_1, \gamma_1, \mu_1), \dots, (\hat{y}_m, \gamma_m, \mu_m)$ and let $(\hat{y}_s, \gamma_s, \mu_s)$ be the prediction with the smallest confidence γ_s . This particular confidence level γ_s is meaningful to us for the following reasons:

1. If we would output prediction sets $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ at significance $\varepsilon = 1 - \gamma_s$, then all these prediction sets would have a cardinality ≤ 1 .
2. Given that $\varepsilon = 1 - \gamma_s$, we expect that the prediction sets $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ (some of which may be empty) contain the true class label with frequency γ_s .
3. If we replace the empty prediction sets in the collection $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ with prediction sets containing any arbitrary label, we still expect the prediction sets to contain the true class label with frequency (at least) γ_s .
4. Given (1–3), we expect the labels $\hat{y}_1, \dots, \hat{y}_m$ to be erroneous with a frequency at most $\varepsilon = 1 - \gamma_s$ and, by extension, to contain on average at most $(1 - \gamma_s)m$ incorrect predictions.

For any arbitrary confidence measure $\gamma_r \in \gamma_1, \dots, \gamma_m$, we may additionally conclude the following:

5. If we would output prediction sets $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ at significance $\varepsilon = 1 - \gamma_r$, then all prediction sets corresponding with confidence $\gamma_i \geq \gamma_r$ have a cardinality ≤ 1 , while all prediction sets corresponding with confidence $\gamma_i < \gamma_r$ have a cardinality ≥ 2 .
6. Given that $\varepsilon = 1 - \gamma_r$, we expect that the prediction sets $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ (some of which may be empty, and some of which may contain two or more labels) contain the true class label with frequency γ_r .
7. If the classification problem is binary, all erroneous predictions lie among the prediction sets corresponding with a confidence $\gamma_i \geq \gamma_r$.
8. If we replace the empty predictions sets in the collection $\Gamma_1^\varepsilon, \dots, \Gamma_m^\varepsilon$ with prediction sets containing any arbitrary label, we still expect the prediction sets to contain the true class label with frequency (at least) γ_r .
9. Given (5–8), we expect the labels $\hat{y}_l \in \hat{y}_1, \dots, \hat{y}_m : \gamma_l \geq \gamma_r$ to be erroneous with a frequency at most $\varepsilon = 1 - \gamma_r$ and, by extension, to contain on average at most $(1 - \gamma_r)m$ incorrect predictions.

Taken together, we use the reasoning described above to define the following prediction procedure:

1. Given a test set $\vec{x}_1, \dots, \vec{x}_k$, produce confidence-credibility predictions $(\hat{y}_1, \gamma_1, \mu_1), \dots, (\hat{y}_k, \gamma_k, \mu_k)$.
2. For each prediction $(\hat{y}_j, \gamma_j, \mu_j)$, compute $\hat{k}_j = k(1 - \gamma_j)$. Here, $\hat{k}_j$ is the estimated number of erroneous predictions among those predictions with confidence $\geq \gamma_j$. Note that $\hat{k}$ is anti-monotonic with respect to γ , i.e., $\gamma_i \geq \gamma_j \rightarrow \hat{k}_i \leq \hat{k}_j$.
3. Output the predictions

$$\{\hat{y}_j \in \{\hat{y}_1, \dots, \hat{y}_k\} : \hat{k}_j \leq \kappa\},$$

where κ is user-specified and denotes the maximum number of errors that we allow on the test set.

This procedure selects a subset $\mathbf{Y}^\kappa$ of all the predictions made, such that $\mathbf{Y}^\kappa$ is the largest possible set that includes at most κ errors on average. Effectively, the procedure filters out low-confidence predictions, while simultaneously estimating the number of erroneous predictions contained within the remaining set of high-confidence predictions.

In Paper V, we provide an empirical assessment of this reject-option conformal classification procedure, generating (for a number of binary classification problems) prediction sets containing either 1, 5 or 10 expected errors. In all cases, the empirical error rate of the output prediction set corresponds well with the supplied κ parameter; however, the rejection rate of predictions depends on the difficulty of the problem. If the classification problem is difficult to model, the conformal classifier generates low-confidence predictions at an increased rate, so our procedure will reject a larger subset of the candidate predictions. In essence, our proposed procedure translates the validity and efficiency qualities of a standard conformal classifier into something that is more readily interpreted by an analyst: efficiency corresponds with the rate of non-rejected predictions, and validity corresponds with the coherence between κ and the true error rate of the non-rejected predictions.

4.1.2 Regression

Compared to conformal classifiers, conformal regression models suffer less in terms of interpretability. In particular, it is much less obvious in a regression setting whether certain predictions should be considered more credible than others (given the same level of confidence). Using a non-normalized conformal regression model, all prediction intervals are of the same size, and without qualifying the assessment further (by considering, e.g., the true label or the object itself), there is thus no apparent reason to trust any one prediction over another. If a normalized conformal regression model is used, the output prediction intervals do indeed vary in size, and we may hypothesize that prediction interval size would correlate with the prediction error probability. However, because the literature has not found any clear such connection, we may just as well hypothesize that the discrete nature of classification problems is a root cause for this particular behavior to arise. With regards to the interpretability of conformal regression models, we thus address a smaller (and perhaps more esoteric) problem although still however, with the intention of reducing the chances of misinterpreting the prediction regions that are output by conformal predictors.

The computational complexity of a conformal classifier depends on the number of possible label assignments. In a transductive conformal classifier, we are required to recompute the underlying model once for every possible label assignment. For both transductive and inductive (as well as aggregated) conformal classifiers, one p -value is computed for each possible label assignment by enumerating the classes in the label space. In a regression setting, this kind of behavior obviously becomes intractable, as there is an infinite number of possible label assignments for the test object. Although there are ways to

circumvent this such as using the k -nearest neighbor algorithm as the underlying model [Papadopoulos et al., 2011], or by realizing that it is sufficient to consider only those label assignments that would cause the p -value to change (information that can be obtained from the calibration set; the number of candidate labels is effectively bound by the size of the calibration set [Papadopoulos et al., 2011]), we can solve the problem more generally by making two specific concessions. First, we ensure that the model is constructed using the inductive mode. Second, we ensure that the nonconformity function used is (at least partially) invertible.

The most common choice of such an invertible nonconformity function is the absolute error measure of the underlying model, previously given in Equation 2.23,

$$\alpha_i = f[\zeta^t, z_i] = |y_i - h(\vec{x}_i)|,$$

where h is the underlying regression model trained on the training set ζ^t .

Given a prediction for a new object, $\hat{y}_{k+1} = h(\vec{x}_{k+1})$, we use the partial inverse to construct the prediction region, as previously given in 2.24, as

$$\Gamma_{k+1}^\varepsilon = h(\vec{x}_{k+1}) \pm \alpha_\delta,$$

where α_δ is a nonconformity score taken from the calibration set, dependent on the specific significance level ε . For a more detailed description of the procedure, see Section 2.5.

Our focus here (and, in Paper I) is the following: by using the nonconformity measure above, and its partial inverse, to construct conformal prediction intervals, we are implicitly making the assumption that the errors made by the underlying model follow some symmetric distribution. Although this implicit assumption should never affect the overall validity of the conformal regression model (prediction regions will still see an error rate ε regardless of the true error distribution of the underlying model), it may severely affect the efficiency of the conformal regression model. In addition, it may lead to predictions that appear misleading given improper interpretation.

To illustrate one issue with this strategy, we assume the following: if we are given a prediction interval (a, b) , given a significance level ε , we may comfortably postulate that the interval (a, b) contains the true label y with probability $1 - \varepsilon$. In certain applications, however, we may not be particularly interested in the upper bound b (e.g., we may want to ensure that a chemical compound is *at least* level- a water-soluble). That is, we know that the likelihood of the true label laying outside the interval (a, b) is ε , although we wish to know the probability that $y < a$. Here, we can only guarantee that the probability that

$y < a$ is, at most, ε . Naturally, the same holds true for the converse statement: the probability that $y > b$ is at most ε .

In Paper I, we introduce an alternative nonconformity measure for regression problems where the sign of the error is kept. That is, we retain the information concerning the underlying model's tendency to over-estimate or under-estimate the true label. We define the signed-error nonconformity function

$$\alpha_i = y_i - \hat{y}_i, \quad (4.2)$$

and re-define the predictive test to emulate two one-tailed tests rather than a single two-tailed test, according to

$$\Gamma_{k+1}^\varepsilon = (\hat{y}_{k+1} + \alpha_l, \hat{y}_{k+1} + \alpha_h), \quad (4.3)$$

where α_l and $\alpha_h \geq \alpha_l$ are, respectively, nonconformity scores taken from the calibration set corresponding to the underlying model's tendency to over-estimate the true label (negative α -values) or under-estimate the true label (positive α -values).

As shown in the empirical results provided in Paper I, the signed-error nonconformity function solves the problem outlined above relating to the standard absolute-error nonconformity function. Since the new prediction region (a, b) follows the error distribution of the underlying model, we are able to provide the following:

- $P(y_{k+1} \notin [a, b]) = \varepsilon$
- $P(y_{k+1} < a) = 0.5\varepsilon$
- $P(y_{k+1} > b) = 0.5\varepsilon$

4.2 Stable and Unstable Nonconformity Functions

In Paper II, we take a closer look at how the choice of nonconformity function affects the efficiency of a conformal predictor. Specifically, we investigate the relationship between *instability* in the underlying learning algorithm and efficiency of the resulting conformal predictor. Here, we consider the notion of model instability described by Breiman [1996], i.e., the tendency of a learning algorithm to substantially modify its learned rule given small differences in the training data.

We show, in Paper II, that unstable learning algorithms such as decision tree learning algorithms are an unsuitable basis for nonconformity functions in

transductive conformal predictors. That is, if the learning algorithm is unstable, then (by definition) a model induced from a training set including the test object $(\vec{x}_{k+1}, \tilde{y})$ will accommodate the test pattern even when mislabeled. In such cases, the misclassified pattern $(\vec{x}_{k+1}, \tilde{y})$ will typically receive a small nonconformity score as an unstable model inferred from a training set including $(\vec{x}_{k+1}, \tilde{y})$ as a training pattern will accurately predict that $\tilde{y}$ is—falsely—the true label of $\vec{x}_{k+1}$. Therefore, rejecting the incorrect label $\tilde{y}$ becomes difficult, resulting in a poor efficiency of the conformal predictor. However, in an inductive conformal predictor, the instability of the underlying learning algorithm largely does not affect efficiency in this manner. Because the mislabeled test object is not included in the training data, there is no apparent connection between the model’s instability and the likelihood of mistakenly producing a good prediction for the pattern $(\vec{x}_{k+1}, \tilde{y})$. Hence, for a transductive conformal predictor we should prefer stable learning algorithms over unstable ones; however, for inductive conformal predictors, this choice becomes less important.

In cases where we wish to construct a conformal prediction ensemble model, learning algorithm stability influences not only efficiency but also validity of the conjugate model (as noted in Paper IV). The details regarding this are discussed in the next section, where we provide a thorough summary of findings relating to conformal predictor ensemble construction.

4.3 Conformal Predictor Ensemble Calibration

When introducing conformal predictor ensembles, as cross-conformal predictors, Vovk [2015] noted that validity is not automatic in general. Under some conditions given by the underlying nonconformity function, it is possible that a conformal predictor constructed using the aggregated conformal prediction schemata (including cross-conformal and bootstrap conformal predictors) does not adhere to the required error level determined by ϵ . However, validity is “violated in a non-interesting way” [Vovk, 2015], as predictions are expected to be overly conservative on average, unnecessarily including incorrect labels in the prediction regions rather than being overly optimistic and therefore rejecting the true label more often than ϵ .

As we reveal in Paper IV, this violation is a possible result of choosing an unstable learning algorithm as the basis for the conformal prediction ensemble. Although learning algorithm instability is generally beneficial for ensemble construction [Breiman, 1996], the case regarding (in)stability becomes essentially self-contradictory with regards to (p -value aggregation-based) conformal predictor ensembles. On the one hand, we want the member models of an en-

semble to show diversity in their behavior (if they all behave identically, we are simply maintaining multiple copies of the same model), which effectively requires some amount of instability in the model inference process. On the other hand, if the model inference process is unstable (i.e., the ensemble member models and the p -values they generate are less than perfectly correlated), it is difficult to find a generally valid p -value aggregation strategy. In Paper IV, we provide a detailed treatment of this issue related to calibrating conformal predictor ensembles, concluding that it may be beneficial to investigate the capabilities of conformal predictor ensembles that do not rely on p -value aggregation, such as the out-of-bag calibrated conformal predictor described in the following section.

4.3.1 Out-of-bag Calibrated Conformal Predictors

Initially proposed by Devetyarov and Nouretdinov [2010], out-of-bag calibration removes the need for leave-one-out calibration in a transductive conformal prediction setting. The out-of-bag calibration strategy was further developed by Johansson et al. [2014], who proposed their use as an alternative method for calibrating inductive conformal predictors based on bagged ensembles (specifically, Random Forests). Boström et al. [2017] provide further refinement of the out-of-bag calibration procedure and give a proof of the procedure’s validity in regression settings. Results presented in this thesis suggest that the proof provided by Boström et al. [2017] may be incorrect and that out-of-bag conformal predictors should likely not be considered automatically valid.

In the above cases, the authors assume that the underlying model is a bagging-enabled ensemble model (specifically, a Random Forest). In Paper VI, however, we provide a generalized definition of the out-of-bag calibration framework that is not specifically bound to models that are inherently based on bagged ensembles.

Similar to the bootstrap conformal predictor, the process for constructing an out-of-bag conformal predictor first draws a series of bootstrap samples $\zeta_1^t, \dots, \zeta_n^t$ from the initial training data ζ^k . From each bootstrap sample ζ_m^t , we induce a classifier h_m that will serve as basis for the nonconformity function.

Any bootstrap sample contains approximately two thirds of the examples contained in the original data, and the examples left out of a particular bootstrap sample are said to be *out-of-bag* for that sample. We denote this out-of-bag property by

$$O_m^i = \begin{cases} 1, & \text{if } z_i \notin \zeta_m^t \\ 0, & \text{otherwise,} \end{cases}$$

i.e., $O_m^i = 1$ if the example z_i was excluded from the bootstrap sample (i.e., it is out-of-bag).

Assuming we define a collection of nonconformity score functions on the form $f[Z_m^t, (\cdot)]$, we compute the aggregated nonconformity scores for the examples in training set as

$$\alpha_i = \frac{1}{\sum_{m=1}^n O_m^i} \sum_{m=1}^n O_m^i \times f[Z_m^t, (\vec{x}_i, y_i)], \quad (4.4)$$

i.e., the nonconformity score of any given training pattern z_i is the average score given by the nonconformity measures for which z_i is out-of-bag.

Finally, we compute the nonconformity score of a test pattern as

$$\alpha_{k+1}^{\tilde{y}} = \frac{1}{\sum_{m=1}^n O_m^r} \sum_{m=1}^n O_m^r \times f[Z_m^t, (\vec{x}_{k+1}, \tilde{y})], \quad (4.5)$$

and the resulting p -value as

$$p_{k+1}^{\tilde{y}} = \frac{\left| \left\{ z_i \in Z^k \setminus \{z_r\} : \alpha_i \geq \alpha_{k+1}^{\tilde{y}} \right\} \right| + 1}{|Z^k \setminus \{z_r\}| + 1}, \quad (4.6)$$

where $r \in U[1, k]$ is the index of a randomly chosen training example.

The empirical results provided in Paper VI show that the out-of-bag calibration procedure yields highly efficient conformal predictor ensembles regardless of the choice of underlying model. Indeed, we obtain good results using both decision trees and support vector machines, models that are at opposite ends of the stability spectrum. Unfortunately, we are unable to prove theoretical validity of the procedure. In fact, we show why it is *not* an automatically valid process, although we simultaneously provide arguments for why we expect the procedure to empirically behave like a valid conformal predictor under realistic circumstances. In particular, we expect the out-of-bag calibration procedure to provide empirically sound predictions if the nonconformity measure used possesses two specific properties: the nonconformity score average decreases (approaches zero) with an increasing training set size (i.e., the underlying model's performance improves with the training set size k); and, as the training set size increases, the performance increase in the underlying model slows down as the training set size increases, i.e., performance of the underlying model is approximately logarithmic with respect to training set size. These two properties are expected from any sufficiently powerful underlying classification algorithm.

4.3.2 Ensemble Validity and Efficiency

In Papers IV and VI, we provide a thorough investigation into conformal predictor ensembles, and focus especially on the effect the nonconformity function has on the observed performance metrics. In these papers, we provide strong evidence that conformal predictor ensemble strategies that rely on p -value aggregation (e.g., averaging) are unsuitable if the underlying learning algorithm is unstable (i.e., highly variant with respect to its training data). Conversely, we may state that unstable nonconformity functions are an unsuitable choice if a p -value aggregation strategy is used.

In terms of efficiency and validity, the distinction between stable and unstable nonconformity functions is quite clear: if the underlying model is unstable, then an aggregated conformal predictor tends to output p -values distributed unimodally rather than uniformly, leading to (conservative) invalidity and reduced efficiency for low (interesting) values of ε . If instead the underlying model is stable, empirical error rates are close to what is expected from a valid conformal predictor, and the efficiency of the aggregated conformal predictor does not deteriorate.

Although out-of-bag conformal predictors, like aggregated conformal predictors, are not provably valid in general, they do (again, like aggregated conformal predictors) show empirically sound error rates. However, unlike ensemble strategies based on p -value aggregation, the efficiency and (empirical) validity of an out-of-bag conformal predictor appears to be independent of the stability of the underlying learning algorithm. Regardless of whether the underlying model is stable or unstable, out-of-bag conformal predictors appear to generally improve efficiency (compared to inductive variants) while simultaneously not producing an excess (or, indeed, dearth) of erroneous predictions.

4.3.3 Ensemble Stability

In addition to our generalized out-of-bag calibration procedure, we propose a new evaluation dimension of conformal predictors that is particularly pertinent in the context of conformal predictor ensemble creation: *p -value stability*. Our notion of stability is related to but distinct from that of Breiman [1996] that we previously leveraged to find suitable underlying learning algorithms for nonconformity functions. Here, we take stability to denote the idea of reproducibility of predictions in the conformal prediction process. We note that a prediction region Γ_{k+1}^ε produced by an inductive conformal predictor depends on the particular partitioning of training and calibration sets, leading to potentially poor reproducibility in predictions (as a result of poor stability in the generated p -values). On repeating the same conformal prediction procedure

on a given test object, only changing the choice of training and calibration set partitions between repetitions, we may obtain contradictory prediction regions (e.g., $\{0\}$ and $\{1\}$) each correct with the same confidence ε .

Empirical results from Paper VI, show that p -value stability in conformal predictor ensembles displays behavior similar to that of efficiency and validity. In an aggregated conformal predictor, p -value stability is improved compared to an inductive conformal predictor when the underlying algorithm is stable. Conversely, p -value stability in an aggregated conformal predictor is decreased compared to an inductive conformal predictor if the underlying algorithm is unstable.

Again, the out-of-bag conformal predictor appears unaffected by stability of the underlying learning algorithm. Although the p -value stability does not necessarily appear to improve (compared to an inductive conformal predictor) when the underlying model is stable, no evidence suggests that p -value stability degrades if the underlying model is unstable.

5. Concluding Remarks

*The music was too loud, the food was cold, the drinks were few,
and the people were many. It was everything I expected, and less.
I'm never going outside again unless I need someplace to throw
up.*

—Bernard Black

5.1 Discussion

In this thesis, our work has primarily been concerned with highlighting and resolving some unclear or unintuitive properties of conformal predictors. Specifically, we provide practical guidance on how to deploy and interpret conformal prediction systems that display appropriate characteristics in their predictions. Below, we relate the findings of the papers included in the thesis to the research questions posed.

How can we provide an interpretation of conformal predictors that does not rely on an understanding of p-value statistics?

We have identified some problems with the interpretability of conformal predictors in both classification and regression settings. Specifically, we noted that correct interpretation of the prediction regions produced by conformal classifiers requires that the user has a deep understanding of p -value statistics. Similarly, we noted that conformal regression models (in the way they are typically created) require the user to have detailed knowledge of the conformal prediction process to avoid erroneous interpretation. In both cases, a lack of detailed understanding of the conformal prediction process may lead to elevated error rates relative to the user's expectations.

For conformal classifiers, we propose to resolve the issue of interpretability by introducing two different means of post-processing the conformal predictions in ways that effectively remove the source of possible misinterpretation. First, we introduce a procedure for adjusting the confidence measures of prediction regions output for binary classification problems so that the reported

expected error rate applies specifically to singleton predictions (i.e., predictions containing exactly one class label), which are typically the predictions of most interest. Second, we devise a post-processing procedure that aims to maximize the refinement rate of the test set by introducing a reject option. Here, our proposed method leverages a binary conformal classifier to output single-label predictions for a subset of the test objects such that the output set contains predictions for as many test objects as possible while simultaneously containing at most κ erroneous predictions, where κ is user-specified.

Several other avenues of research address the potential issues of interpretability in conformal predictors, and related frameworks, in different ways. A theoretical framework has been proposed for leveraging conformal predictive distributions and utility functions in decision making [Vovk and Bendtsen, 2018; Vovk et al., 2017], but the practical usefulness of the framework is yet to be tested [Vovk and Bendtsen, 2018]. More recently, a line of research proposes replacing the p -values of conformal predictors with e -values [Vovk and Wang, 2019a,b, 2020], leading to an effect reminiscent to that proposed in our classifier with a reject option, where (non-relative) error frequency is controlled, rather than error probability. Although publications on conformal predictors with e -values primarily address topics relating to model combination and multiple testing problems, the use of e -values appears to be a potential contender for providing easier-to-interpret conformal prediction models. Finally, the Venn prediction framework [Vovk and Petej, 2014] describes a calibration procedure very similar to conformal prediction, where confidence assertions are made in the form of well-calibrated probabilities rather than p -value statistics.

Although promising, the approaches of transforming conformal classification models through the use of utility functions or p -values are still novel and present potential short-term risks to interpretability and usability before the frameworks are fully matured. By contrast, our methods for re-calibrating error frequency estimates (either relative or non-relative) present predictions in a simpler form—a collection of single-label predictions coupled with an expected error frequency. The obvious disadvantage of our strategies is that they are applicable only in batch testing scenarios, where the entire unlabeled test set is available beforehand.

For regression problems, we propose a novel signed-error nonconformity measure, that retains more information about the error distribution of the underlying model than the previously available absolute-error nonconformity measure commonly used for conformal regression models. Like conformal regression models based on the absolute-error nonconformity function, our proposed conformal regressor provides predictions informing the user that y_{k+1} lies on the

interval $[a, b]$ with probability $1 - \varepsilon$. Unlike absolute-error conformal predictors, the prediction regions output by our proposed conformal classifier additionally have the property that the probability that y_{k+1} is less than a , or greater than b , are both exactly $\frac{\varepsilon}{2}$.

The conformal predictive distribution framework [Vovk et al., 2017] likely represents the biggest step toward improving the applicability of conformal regression models, as they can produce posterior predictive probability distributions rather than simple (ε -bound) prediction intervals. From our perspective, the procedure for conducting balanced one-tailed calibration complements the work on conformal predictive distributions while providing a means for improving the prediction intervals produced by ordinary conformal regression models.

How does the choice of nonconformity function affect the efficiency, validity and interpretability of a conformal predictor?

Throughout our work, we have repeatedly made reference to the notion of stable and unstable learning algorithms [Breiman, 1996] to denote algorithms whose learned prediction rules have a small (or great) dependency on the particular training examples used to induce the model. In all cases where we have investigated the relationship between stability of the underlying learning algorithm in relation to properties of the resulting conformal predictor, we have found that stable learning algorithms generally serve as a more suitable basis for nonconformity measures compared to unstable alternatives.

In transductive conformal classifiers, nonconformity measures based on unstable learning algorithms may have a detrimental effect on informational efficiency. Since the test object is included in the training set used to induce the underlying model, the label tentatively assigned to the test object may substantially affect the prediction rule learned by the underlying model. That is, if the test object is tentatively assigned an erroneous label, the underlying model may (incorrectly) provide an accurate prediction of the test object, leading to a small p -value for the erroneous label, which is then unnecessarily included in the resulting prediction region.

In the most common type of conformal prediction ensemble (i.e., aggregated conformal predictors and their variants), unstable underlying learning algorithms appear to have an even greater negative effect (relative to stable underlying learners). Here, efficiency is negatively affected in a similar fashion (although due to other underlying mechanisms) as in a transductive conformal predictor. Simultaneously, the empirical error rate of an aggregated conformal predictor deviates from the desired confidence level at an increasing rate when

the underlying model is excessively random.

For inductive conformal predictors and alternative conformal prediction ensemble strategies such as the out-of-bag conformal predictor, we have not been able to establish any clear relationship between underlying model stability and properties of the resulting conformal predictor. These particular conformal prediction procedures seem less sensitive to the nature of the nonconformity measure, and appear to consistently provide prediction regions with predictable characteristics regardless of the choice of nonconformity function.

To our knowledge, the work presented in this thesis so far is the only research conducted studying the relationship between fundamental properties of the underlying learning algorithm and the resulting conformal predictor.

How can informationally efficient and statistically valid ensembles be created from conformal predictors?

Our efforts to provide guidance on conformal prediction ensemble creation are two-fold. First, we investigate the behavior of available common-use conformal prediction ensemble methods (i.e., variants of aggregated conformal predictors). Second, we propose an alternative general method for conformal prediction ensemble construction that attempts to alleviate the weaknesses identified in aggregated conformal predictors.

As previously noted, we have established that the choice of nonconformity function may substantially affect the behavior of aggregated conformal predictors in such a way that both efficiency and validity may suffer as the result of selecting an inappropriate nonconformity function. As an alternative, we provide a generalization of the out-of-bag calibration procedure for conformal prediction ensemble training that does not show any particular dependency on properties of the underlying learning algorithm but displays favorable empirical efficiency and validity results regardless of the choice of nonconformity measure.

Unfortunately, much like the aggregated conformal prediction procedure, we are unable to show any general theoretical validity for the proposed out-of-bag calibration procedure. Instead, we rely on empirically obtained results, that suggest that the proposed out-of-bag conformal predictor ensemble generally behaves like a provably valid inductive conformal predictor in terms of the observed error rate, while simultaneously providing significant improvements in informational efficiency.

Over the last few years, multiple alternatives have been proposed to the standard cross-conformal prediction strategy, displaying different characteristics

and levels of success. Balasubramanian et al. [2015] study several alternative p -value combination methods, none of which seem able to produce valid conformal predictors in general. In fact, the proposed methods appear to at least occasionally violate validity in such a way that the error frequency is substantially elevated from the expectation. Similarly, the majority vote combination strategy proposed by Cherubin [2019] does not appear to produce generally valid conformal predictors but requires that the combined models are independent for theoretical results to hold. Interestingly, the behavior of the majority vote ensembles appears opposite to that of a cross-conformal predictor, which effectively requires that the models are strongly dependent in order to maintain empirical calibration. The synergy conformal prediction strategy [Gauraha and Spjuth, 2018] appears to maintain theoretical validity while combining multiple predictive models. However, as only a single calibration set is used throughout the strategy, it does not appear that the synergy conformal prediction strategy is necessarily distinct from an inductive conformal predictor using an ensemble model as the underlying predictor. Likewise, the conformal predictor combination strategy proposed by Toccaceli and Gammerman [2017, 2019] requires an additional calibration step, and its construction appears effectively identical to that of a synergy conformal predictor using an ensemble of p -value predictors as the basis for the nonconformity function. Finally, e -values appear to present an opportunity for combining conformal predictors in a way superior to p -values, by maintaining validity even through simple averaging [Vovk, 2020; Vovk and Wang, 2019b]. Naturally, this requires an adoption of conformal e -predictors over conformal p -predictors in order to prove generally useful.

Although our proposed strategy of out-of-bag calibration is not generally valid, it is evident from related research that finding a single all-purpose ensemble strategy for conformal prediction remains difficult. Thus, we consider our contribution important as it provides additional theoretical insights into conformal prediction ensemble design and an empirically competitive conformal predictor combination strategy.

5.2 Conclusions

As with many—perhaps all—technical doctoral theses, the subject matter presented in these pages may strike the reader as excessively esoteric. As a pragmatically disposed individual, I would like to dedicate this section of the thesis to its true purpose and provide some pieces of practical guidance on the design and use of conformal predictors, one village idiot to another.

A satirical summary of our findings may read as follows: “To maximize efficiency, use a constant function as the nonconformity measure. To maximize validity, do not observe the predictions after they are made. And, don’t make ensembles.” In unpacking this piece of self-amusement, I hope to satisfactorily convey the key points presented in this thesis and provide the reader with some guidance on how a conformal predictor may be configured to retain the expected properties of such a predictive model—efficiency and validity—as well as some insight into how to properly interpret the output from such a predictive model.

To maximize efficiency, use a constant function as the nonconformity measure.

By now, the number of times we have referenced Leo Breiman [1996] and his notion of (un)stable learning algorithms appears to approach infinity at a rapid pace; thus, it seems only natural that a significant portion of our conclusions should be dedicated to once more illustrate the relationship between learning algorithm stability, nonconformity functions, and conformal predictor validity and efficiency. In simple terms, our results (both theoretical and empirical) suggest that unstable learning algorithms, i.e., those classification and regression algorithms that learn significantly different prediction rules given only relatively small changes in training data, are not suitable as a basis for nonconformity functions in conformal predictors. In particular, transductive conformal predictors and aggregated conformal predictor ensembles (including cross-conformal predictors and bootstrap conformal predictors) systematically show poor performance, in terms of efficiency (prediction region size) when an unstable underlying learning algorithm is used. In addition, the validity (error rate) of such aggregated conformal predictor ensembles may suffer in such a way that the error probability of the generated prediction regions becomes less predictable than normally anticipated.

Efficiency and validity of inductive conformal predictors and out-of-bag calibrated conformal predictor ensembles do not, however, appear to be affected by the stability of the underlying learning algorithm to any large extent. If an unstable learning algorithm, such as decision tree induction, is desired, it appears most appropriate to construct the conformal predictor using any of these

two procedures.

We consider the results presented in this thesis to provide much-needed guidance on the appropriate selection of nonconformity functions in several different conformal prediction contexts, in particular regarding the choice of underlying machine learning model.

To maximize validity, do not observe the predictions after they are made.

It is perhaps not surprising that the earliest papers included in this thesis are dedicated to attempts at providing comprehensible interpretations of the prediction regions generated by conformal predictors just as it is most likely not just happenstance that several instances of Randall Munroe’s XKCD webcomic are dedicated to humorous misrepresentations of p -value statistics.

The output obtained from a conformal classifier is potentially confusing to anyone not intimately familiar with p -value statistics. Given an object, the conformal classifier provides a set of possible class assignments together with the promise that one of these class assignments is correct with probability $1 - \epsilon$. Although the conformal classifier unequivocally holds its promise, we must take care in how we use the prediction set or we may inadvertently elevate the error probability of (our interpretation of) the prediction. If a prediction region contains the full set of possible class assignments or is empty, we may easily be inclined to simply disregard the prediction as non-informative. However, this disregard additionally requires us to adjust our expectations (in terms of error probability) for successive predictions, lest we increase the false discovery rate for future events.

To alleviate the onus on the user of a conformal predictor to correctly interpret its predictions, this thesis provides two post-processing procedures for conformal prediction regions, which hide the potentially confusing p -value statistics from the user. In both cases, we focus on providing actionable information regarding those most directly interesting prediction regions, i.e., those containing only a single class label. The first proposed method provides the user with the error probability of singleton prediction regions. The second proposed method provides the user with a collection of singleton prediction regions with a predefined (non-relative) error frequency.

In recent years, research addressing the question of interpretability of conformal prediction has increased, suggesting there is a need to provide alternative means of presenting the outputs of conformal predictor. While we welcome any efforts to address this issue, we believe that dissemination of recently proposed methods to a broader audience might take at least several years. To help this progress, we believe our contributions provide an appropriate stepping-

stone until alternatives are fully developed and communicated.

And, don't make ensembles.

Many of the conformal prediction ensemble strategies discussed in this thesis share one interesting characteristic: neither is provably valid in a statistical sense. Although this inconvenient fact may suggest that the two methods are of little use, we (as pragmatic individuals) think nothing of the sort; rather, while conceivably we may be coerced into admitting that neither of these methods generate proper conformal predictors (which are, after all, automatically valid by definition), most strategies act, empirically, much in the same way that we would wish from a conformal predictor ensemble.

Although not generally valid, aggregated conformal predictors are expected to behave in a conservatively valid manner for low (interesting) values of ϵ . That is, given a realistic choice of ϵ , the empirical error rate is not expected to exceed the desired level. Although this may not appear immediately problematic, this typically leads to unnecessarily large prediction regions, an effect that is exacerbated if the underlying learning algorithm is unstable. Therefore, for any aggregated conformal predictor variant it seems appropriate to use only stable underlying learning algorithms as a basis for the nonconformity function; however, even then, informational efficiency may suffer.

Although we are unable to show any, even conservative or approximate, theoretical validity for the out-of-bag conformal predictor ensemble strategy we propose, empirical results are promising as error rates consistently adhere to the desired ϵ at the same rate as an inductive conformal predictor. In addition, prediction regions generally appear smaller than those generated by both inductive and aggregated conformal predictors, regardless of nonconformity function used. Hence, theoretical validity aside, we deem out-of-bag calibration to be a serious contender as a generally appropriate conformal predictor ensemble strategy.

5.3 Future Directions

In this thesis we have thoroughly investigated what impact machine learning algorithm stability has on nonconformity functions and conformal predictors based on them. For a conformal predictor to behave satisfactorily in all regards, it is often necessary to consider this particular property of the underlying algorithm. Naturally, we are inclined to believe that there exist other properties of such underlying algorithms that may have a substantial and predictable effect on conformal predictor behavior. For example, we have hinted at (but not made any further investigation into) the fact that model ranking performance (rather than model accuracy, e.g.) may be a key capability for successful nonconformity functions. We consider the examination of the relationship between conformal predictor performance and other such qualities of the underlying model an important task, as the knowledge of such relationships guides us in designing conformal predictors with increasingly predictable behavior. In addition, a deeper understanding of what constitutes a good nonconformity function (conditionally or unconditionally) may open up new interesting alternatives for nonconformity function design, beyond those based on traditional classification and regression algorithms.

We have contributed to the understanding of the most common type of conformal predictor ensemble, the aggregated conformal predictor, including cross-conformal predictors and bootstrap conformal predictors. In addition, we have developed a general out-of-bag calibration strategy for conformal predictor ensemble construction that alleviates some, but not all, problems associated with aggregated conformal predictors. Since the initiation of the work presented in this thesis several promising proposals have been made for alternative methods of conformal prediction ensemble construction. We believe that a consolidation of ideas relating to conformal prediction ensemble creation is necessary for producing strategies that are consistently able to provide efficient and provably valid predictions. Additionally, we consider it necessary to further assess such promising ensemble methods, to illustrate any inherent strengths and weaknesses compared to available alternatives.

An intuitively accessible interpretation of conformal predictors, in particular conformal classifiers, is a necessary development for their practical use in decision making processes. Here, we suggest alternative interpretations of the output from conformal classifiers, that provide more actionable decision support. We present a method for re-calibrating the expected error frequency specifically on prediction regions containing exactly one class label, and another for performing classification with reject option using conformal prediction. The methods we propose have some apparent downsides: neither is designed for

an online setting but require that the full unlabeled test set be available beforehand, and our re-calibrating method’s expected error frequencies of singleton predictions are only applicable to binary classification problems. Naturally, extending these procedures may result in more general applicability. We expect that the first method (and perhaps the second) could be extended to an online setting by computing relevant statistics on the calibration set rather than the test set; however, care must be taken to avoid violating conditions necessary for maintaining validity. We expect that extending the first method to multi-class problems to be a more difficult task, but consider it potentially possible by imposing restrictions either on the problem or on the underlying learning algorithm. Alternatively, clever use of nonconformity measure conditionality may present an opportunity.

In this work, we have adopted a rather technical interpretation of the idea of conformal predictor usability and interpretability, and see great potential benefit in further studying the strengths and weaknesses of the conformal prediction framework, including the contributions developed in this thesis, through a user-focused lens.

Separate efforts have been made to connect conformal classifiers to more readily interpretable decision bases, such as the work done regarding e -values and conformal predictive distributions. We believe further work in this regard is necessary. One promising approach may be to more closely investigate the relationship between conformal classifiers and Venn predictors.

Sammanfattning

Avhandlingen behandlar området *conformal prediction*, som beskriver en familj prediktiva modeller vars prediktioner associeras med ett konfidensmått, som låter modellerna själva uttrycka sig om sin egen tillförlitlighet. I högriskapplikationer, där dåliga prediktioner kan få allvarliga ekonomiska konsekvenser, eller leda till personskada, tycks en sådan inbyggd säkerhetskontroll högst värdefull, om inte nödvändig. Medan den teoretiska grunden till conformal prediction lades för cirka 20 år sedan, är forskningsområdet fortfarande relativt ungt, och många öppna frågor kvarstår gällande design och användning av conformal prediction-system. I avhandlingen behandlas några av dessa öppna frågor, och fokus läggs på tre specifika karakteristika hos conformal-prediktorer. Först behandlas de så kallade icke-konformitetsfunktionerna (*non-conformity functions*) som ligger till grund för conformal prediction, och sambandet utforskas mellan egenskaper hos icke-konformitetsfunktionerna och de resulterande prediktorerna. Även egenskaper hos de prediktioner som produceras i en conformal predictor undersöks, och två post-processeringsmetoder presenteras i ett försök att bistå med en mer intuitivt begriplig tolkning av dessa prediktioner. Slutligen utforskas strategier för konstruktion av ensembler av conformal prediction-modeller, där svagheter illustreras i vedertagna strategier, följt av en presentation av en ny ensemblestrategi som ämnar adressera dessa svagheter.

References

- Jesús Alcalá-Fdez, Alberto Fernández, Julián Luengo, Joaquín Derrac, Salvador García, Luciano Sánchez, and Francisco Herrera. Keel data-mining software tool: data set repository, integration of algorithms and experimental analysis framework. *Journal of Multiple-Valued Logic & Soft Computing*, 17, 2011. 31
- Arthur Asuncion and David Newman. Uci machine learning repository, 2007. 31
- Laura Badenes-Ribera, Dolores Frías-Navarro, Héctor Monverde-i Bort, and Marcos Pascual-Soler. Interpretation of the p value: A national survey study in academic psychologists from Spain. *Psicothema*, pages 290–295, 2015. 11
- Vineeth N Balasubramanian, Shayok Chakraborty, and Sethuraman Panchanathan. Conformal predictions for information fusion. *Annals of Mathematics and Artificial Intelligence*, 74(1-2):45–65, 2015. 57
- Henrik Boström, Henrik Linusson, Tuve Löfström, and Ulf Johansson. Evaluation of a variance-based nonconformity measure for regression forests. In *Symposium on Conformal and Probabilistic Prediction with Applications*, pages 75–89. Springer, 2016.
- Henrik Boström, Henrik Linusson, Tuve Löfström, and Ulf Johansson. Accelerating difficulty estimation for conformal regression forests. *Annals of Mathematics and Artificial Intelligence*, 81(1-2):125–144, 2017. 49
- Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996. 34, 37, 47, 48, 51, 55, 58
- Leo Breiman, Jerome Friedman, Richard Olshen, and Charles Stone. Classification and regression trees. wadsworth int. *Group*, 37(15):237–251, 1984. 34
- Ioan Buciuc, Constantine Kotropoulos, and Ioannis Pitas. Demonstrating the stability of support vector machines for classification. *Signal Processing*, 86(9):2364–2380, 2006. 34
- Lars Carlsson, Martin Eklund, and Ulf Norinder. Aggregated conformal prediction. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 231–240. Springer, 2014. 12, 26
- Giovanni Cherubin. Majority vote ensembles of conformal predictors. *Machine Learning*, 108(3):475–488, 2019. 57
- Giovanni Cherubin, Ilia Nouretdinov, Alexander Gammerman, Roberto Jordaney, Zhi Wang, Davide Papini, and Lorenzo Cavallaro. Conformal clustering and its application to botnet traffic. In *International Symposium on Statistical Learning and Data Sciences*, pages 313–322. Springer, 2015. 10, 15
- David Colquhoun. An investigation of the false discovery rate and the misinterpretation of p-values. *Royal Society open science*, 1(3):140216, 2014. 11
- David Colquhoun. The reproducibility of research and the misinterpretation of p-values. *Royal society open science*, 4(12):171085, 2017. 11

- Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine learning*, 20(3):273–297, 1995. 34
- Larry G Daniel. Statistical significance testing: A historical overview of misuse and misinterpretation with implications for the editorial policies of educational journals. *Research in the Schools*, 5(2):23–32, 1998. 11
- Dmitry Devetyarov and Ilia Nouretdinov. Prediction with confidence based on a random forest classifier. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 37–44. Springer, 2010. 12, 49
- Dmitry Devetyarov, Ilia Nouretdinov, Brian Burford, Stephane Camuzeaux, Aleksandra Gentry-Maharaj, Ali Tiss, Celia Smith, Zhiyuan Luo, Alexey Chervonenkis, Rachel Hallett, et al. Conformal predictors in early diagnostics of ovarian and breast cancers. *Progress in Artificial Intelligence*, 1(3):245–257, 2012. 10
- Martin Eklund, Ulf Norinder, Scott Boyer, and Lars Carlsson. Application of conformal prediction in qsar. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 166–175. Springer, 2012. 10
- Martin Eklund, Ulf Norinder, Scott Boyer, and Lars Carlsson. The application of conformal prediction to the drug discovery process. *Annals of Mathematics and Artificial Intelligence*, 74(1-2):117–132, 2015. 10
- Alex Gammerman, Ilia Nouretdinov, Brian Burford, Alexey Chervonenkis, Vladimir Vovk, and Zhiyuan Luo. Clinical mass spectrometry proteomic diagnosis by conformal predictors. *Statistical applications in genetics and molecular biology*, 7(2), 2008. 10
- Alexander Gammerman, Volodya Vovk, and Vladimir Vapnik. Learning by transduction. In *Proceedings of the Fourteenth conference on Uncertainty in artificial intelligence*, pages 148–155. Morgan Kaufmann Publishers Inc., 1998. 20
- Niharika Gauraha and Ola Spjuth. Synergy conformal prediction. 2018. 57
- Steven Goodman. A dirty dozen: twelve p-value misconceptions. In *Seminars in hematology*, volume 45, pages 135–140. Elsevier, 2008. 11
- Ulf Johansson, Henrik Boström, and Tuve Löfström. Conformal prediction using decision trees. In *2013 IEEE 13th international conference on data mining*, pages 330–339. IEEE, 2013a. 12, 35, 38
- Ulf Johansson, Rikard König, Tuve Löfström, and Henrik Boström. Evolved decision trees as conformal predictors. In *2013 IEEE Congress on Evolutionary Computation*, pages 1794–1801. IEEE, 2013b. 12, 38
- Ulf Johansson, Henrik Boström, Tuve Löfström, and Henrik Linusson. Regression conformal prediction with random forests. *Machine Learning*, 97(1-2):155–176, 2014. 38, 49
- Ulf Johansson, Henrik Linusson, Tuve Löfström, and Henrik Boström. Interpretable regression trees using conformal prediction. *Expert systems with applications*, 97:394–404, 2018. 38
- Ulf Johansson, Tuve Löfström, Henrik Boström, and Cecilia Sönströd. Interpretable and specialized conformal predictors. In *Conformal and Probabilistic Prediction and Applications*, pages 3–22, 2019. 38
- Douglas H Johnson. The insignificance of statistical significance testing. *The journal of wildlife management*, pages 763–772, 1999. 11
- Venkateswara Rao Kagita, Arun K Pujari, Vineet Padmanabhan, Sandeep Kumar Sahu, and Vikas Kumar. Conformal recommender system. *Information Sciences*, 405:157–174, 2017. 15

- Antonis Lambrou, Harris Papadopoulos, and Alex Gammerman. Reliable confidence measures for medical diagnosis with evolutionary algorithms. *IEEE Transactions on Information Technology in Biomedicine*, 15(1):93–99, 2011. 10
- Rikard Laxhammar and Göran Falkman. Conformal prediction for distribution-independent anomaly detection in streaming vessel data. In *Proceedings of the first international workshop on novel data stream pattern mining techniques*, pages 47–55. ACM, 2010. 10, 15
- Andy Liaw, Matthew Wiener, et al. Classification and regression by randomforest. *R news*, 2(3):18–22, 2002. 34
- Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Efficiency comparison of unstable transductive and inductive conformal classifiers. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 261–270. Springer, 2014a.
- Henrik Linusson, Ulf Johansson, and Tuve Löfström. Signed-error conformal regression. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 224–236. Springer, 2014b.
- Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Reliable confidence predictions using conformal prediction. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 77–88. Springer, 2016.
- Henrik Linusson, Ulf Norinder, Henrik Boström, Ulf Johansson, and Tuve Löfström. On the calibration of aggregated conformal predictors. In *Conformal and probabilistic prediction and applications*, pages 154–173. PMLR, 2017.
- Henrik Linusson, Ulf Johansson, Henrik Boström, and Tuve Löfström. Classification with reject option using conformal prediction. In *Pacific-Asia Conference on Knowledge Discovery and Data Mining*, pages 94–105. Springer, 2018.
- Henrik Linusson, Ulf Johansson, and Henrik Boström. Efficient conformal predictor ensembles. *Neurocomputing*, 397:266–278, 2020.
- Tuve Löfström, Henrik Boström, Henrik Linusson, and Ulf Johansson. Bias reduction through conditional conformal prediction. *Intelligent Data Analysis*, 19(6):1355–1375, 2015. 27
- Rupert G Miller. The jackknife-a review. *Biometrika*, 61(1):1–15, 1974. 10
- Radford M Neal. Assessing relevance determination methods using delve. *Nato Asi Series F Computer And Systems Sciences*, 168:97–132, 1998. 31
- Ulf Norinder, Lars Carlsson, Scott Boyer, and Martin Eklund. Introducing conformal prediction in predictive modeling, a transparent and flexible alternative to applicability domain determination. *Journal of chemical information and modeling*, 54(6):1596–1603, 2014. 10
- Iliia Nouruddinov, Sergi G Costafreda, Alexander Gammerman, Alexey Chervonenkis, Vladimir Vovk, Vladimir Vapnik, and Cynthia HY Fu. Machine learning classification with confidence: application of transductive conformal predictors to mri-based diagnostic and prognostic markers in depression. *Neuroimage*, 56(2):809–813, 2011. 10
- Harris Papadopoulos. Inductive conformal prediction: Theory and application to neural networks. In *Tools in artificial intelligence*. InTech, 2008. 12, 24, 34
- Harris Papadopoulos, Kostas Proedrou, Volodya Vovk, and Alex Gammerman. Inductive confidence machines for regression. In *European Conference on Machine Learning*, pages 345–356. Springer, 2002. 22, 24
- Harris Papadopoulos, Alex Gammerman, and Volodya Vovk. Reliable diagnosis of acute abdominal pain with conformal prediction. *Engineering Intelligent Systems*, 17(2):127, 2009. 10

- Harris Papadopoulos, Vladimir Vovk, and Alexander Gammerman. Regression conformal prediction with nearest neighbours. *Journal of Artificial Intelligence Research*, 40:815–840, 2011. 12, 46
- Harris Papadopoulos, Nestoras Georgiou, Charalambos Eliades, and Andreas Konstantinidis. Android malware detection with unbiased confidence guarantees. *Neurocomputing*, 280:3–12, 2018. 10
- F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. 34
- Arthur L. Samuel. Some studies in machine learning using the game of checkers. *IBM Journal of research and development*, 3(3):210–229, 1959. 9
- Craig Saunders, Alexander Gammerman, and Volodya Vovk. Transduction with confidence and credibility. 1999. 20
- Robert A Stine. Bootstrap prediction intervals for regression. *Journal of the American Statistical Association*, 80(392):1026–1031, 1985. 10
- Fredrik Svensson, Ulf Norinder, and Andreas Bender. Modelling compound cytotoxicity using conformal prediction and pubchem hts data. *Toxicology research*, 6(1):73–80, 2017. 10
- Fredrik Svensson, Natalia Aniceto, Ulf Norinder, Isidro Cortes-Ciriano, Ola Spjuth, Lars Carlsson, and Andreas Bender. Conformal regression for quantitative structure–activity relationship modeling—quantifying prediction uncertainty. *Journal of Chemical Information and Modeling*, 58(5):1132–1140, 2018. 10
- Paolo Toccaceli and Alexander Gammerman. Combination of conformal predictors for classification. In *Conformal and Probabilistic Prediction and Applications*, pages 39–61, 2017. 57
- Paolo Toccaceli and Alexander Gammerman. Combination of inductive mondrian conformal predictors. *Machine Learning*, 108(3):489–510, 2019. 57
- Paolo Toccaceli, Ilia Nouretdinov, and Alexander Gammerman. Conformal predictors for compound activity prediction. In *Symposium on Conformal and Probabilistic Prediction with Applications*, pages 51–66. Springer, 2016. 10
- Leslie G Valiant. A theory of the learnable. *Communications of the ACM*, 27(11):1134–1142, 1984. 10
- Vladimir Vovk. Conditional validity of inductive conformal predictors. In *Asian conference on machine learning*, pages 475–490, 2012. 27
- Vladimir Vovk. Cross-conformal predictors. *Annals of Mathematics and Artificial Intelligence*, 74(1-2): 9–28, 2015. 12, 25, 26, 32, 48
- Vladimir Vovk. Cross-conformal e-prediction. *arXiv preprint arXiv:2001.05989*, 2020. 57
- Vladimir Vovk and Claus Bendtsen. Conformal predictive decision making. In *Conformal and Probabilistic Prediction and Applications*, pages 52–62. PMLR, 2018. 54
- Vladimir Vovk and Ivan Petej. Venn-abels predictors. In *Proceedings of the Thirtieth Conference on Uncertainty in Artificial Intelligence*, pages 829–838. AUAI Press, 2014. 54
- Vladimir Vovk and Ruodu Wang. True and false discoveries with e-values. *arXiv preprint arXiv:1912.13292*, 2019a. 54
- Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination, and applications. *Forthcoming in the Annals of Statistics*, 2019b. 54, 57

- Vladimir Vovk and Ruodu Wang. True and false discoveries with independent e-values. *arXiv preprint arXiv:2003.00593*, 2020. 54
- Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic learning in a random world*. Springer Science & Business Media, 2005. 9, 17, 18, 20, 22, 24, 27
- Vladimir Vovk, Valentina Fedorova, Ilia Nouretdinov, and Alexander Gammerman. Criteria of efficiency for conformal prediction. In *Symposium on Conformal and Probabilistic Prediction with Applications*, pages 23–39. Springer, 2016. 35
- Vladimir Vovk, Jieli Shen, Valery Manokhin, and Min-ge Xie. Nonparametric predictive distributions based on conformal prediction. In *Conformal and Probabilistic Prediction and Applications*, pages 82–102. PMLR, 2017. 54, 55
- Harry Wechsler. Cyberspace security using adversarial learning and conformal prediction. *Intelligent Information Management*, 7(04):195, 2015. 10